Estimating Climate Change Risk Exposure: a probabilistic approach

Camilo Saldarriaga
ESCP Business School
jcamsalda@gmail.com

Jesse Grabowski
Paris-1 Panthéon-Sorbonne
jessegrabowski@gmail.com

Jan Rielaender
OECD Development Center
Jan.RIELAENDER@oecd.org

Abstract

This paper develops a methodology to quantify countries' exposure to climate change–related disasters, focusing on hydrometeorological events such as floods and storms. We introduce the Probabilistic Geospatial Risk (PGR) model, a fully Bayesian, multi-level system integrating time series, frequency, damage, and spatial sub-models. The framework captures uncertainty across the entire distribution of disaster frequency and damages, enabling robust estimation of both mean and tail risks. Using data from the EM-DAT disaster database, the World Bank, GPCC, and NOAA, we estimate disaster probabilities and damage distributions as functions of climate and development indicators. We further apply a Hilbert Space Gaussian Process (HSGP) to model fine-grained geospatial patterns of disaster occurrence within countries, complemented by state-space projections of climate and economic covariates. The model is demonstrated through two case studies—Costa Rica and the Lao People's Democratic Republic. The resulting geolocalized damage curves and return-year estimates provide a transparent, reproducible tool for assessing future exposure to climate-related risks. This probabilistic approach advances the literature by integrating uncertainty quantification, geospatial modeling, and economic impact estimation into a coherent framework suitable for policy analysis and climate adaptation planning.

JEL Classification: Q54, Q51, C11, C150, C53, Q56, and D81

Keywords: Climate change risk, Bayesian modeling, hydrometeorological disasters, geospatial modeling, probabilistic risk assessment, damage curves, uncertainty quantification.

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1 Introduction

The current paper develops a methodology to quantify countries' exposure to climate change-related disasters, and presents 2 examples of its application for two different countries: Costa Rica and Lao People's Democratic Republic. The methodology is implemented by one joint model called **Probabilistic Geospatial Risk (PGR) model**.

The methodology focuses on evaluating natural disasters as an important channel through which climate change impacts countries' economies. It employs a set of quantitative and probabilistic methods to estimate and predict the impact of these events.

The approach presented here advances the literature in three key ways. First, it employs a Bayesian modeling framework that enables flexible specification of probability distributions, estimation of the full distribution of disaster occurrences and damages—not just mean values—and rigorous quantification of uncertainty. Second, it incorporates a geospatial approach to connect the geographic patterns of disaster events with global climate trends. Third, it integrates several interrelated models, including time series projections of economic and climate variables, estimates of disaster frequency and severity, and spatial modeling of disaster locations. These models come together to produce probabilistic damage curves, quantifying the range of likely disaster damages in a given location, at a given time while propagating uncertainty between models.

Each step is fully Bayesian, allowing incorporation of prior knowledge held by relevant stakeholders. In addition, the structure of the model allows for scenario-based investigation. For example, it allows the incorporation of different climate forecast scenarios. The result is an end-to-end analysis pipeline that transparently generates, for a given country, region, or city, damage distributions fully characterizing tail risks faced by that region. This comprehensive perspective offers a distinct advantage over existing studies by providing a more detailed and nuanced understanding of the risk of climate-related disasters.

2 Literature Review

2.1 CO_2, climate change and disasters

Over the past few decades, mounting scientific evidence has established a link between anthropogenic CO_2 emissions, global warming, and the increasing frequency and severity of natural disasters worldwide. Much of this evidence has been synthesized by the Intergovernmental Panel on Climate Change (IPCC), a scientific body formally mandated to assess climate change based on the latest scientific, technical, and socio-economic information. The IPCC's comprehensive assessments, including the Sixth Assessment Report Intergovernmental Panel on Climate Change (IPCC) (2023) and earlier publications, demonstrate how greenhouse gas (GHG) emissions, primarily from fossil fuel combustion, have driven unprecedented atmospheric temperature increases (Intergovernmental Panel on Climate Change (IPCC), 2021, 2022*a*,*b*). These temperature changes, in turn, have intensified climate-related hazards, including heatwaves, storms, floods, and droughts, all of which pose growing risks to human societies and ecosystems.

The existing literature indicates a clear causal relationship between CO_2 emissions and global warming, a phenomenon well-documented in the Physical Science Basis of the AR6 (Intergovernmental Panel on Climate Change (IPCC), 2021). Global surface temperatures have risen by approximately 1.1°C above pre-industrial levels, with CO_2 identified as the dominant driver. This warming alters the Earth's energy balance, accelerates ice melt, disrupts precipitation patterns, and amplifies the intensity of extreme weather events. The Special Report on Emissions Scenarios (Nakicenovic et al., 2000) lay the groundwork for understanding how different future emission trajectories could lead to distinct climate outcomes, emphasizing the role of mitigation in curbing long-term disaster risks.

Building on this scientific foundation, the AR6 Working Group II Report (Intergovernmental Panel on Climate Change (IPCC), 2022*a*) highlights the growing vulnerability of human and natural systems to climate-related disasters. The report details how warming has increased the exposure of populations and infrastructure to climate hazards and stresses the compounding nature of these risks, especially in low-income regions. It also notes that socioeconomic disparities influence adaptive capacity, exacerbating disaster impacts in vulnerable communities. The way in which climate change has amplified the frequency and magnitude of natural disasters has also been documented by multiple scientific and institutional findings as indicated by Van Aalst (2006).

Finally, the AR6 Working Group III Report (Intergovernmental Panel on Climate Change (IPCC), 2022*a*) underlines the direct link between CO_2 and global warming. The report provides evidence that immediate and sustained emission reductions across all sectors are essential to staying below the 1.5°C or even 2°C warming thresholds established under the Paris Agreement. Delaying mitigation efforts increases the likelihood of irreversible climate tipping points, such as permafrost thawing or coral reef collapse, which could further exacerbate natural disasters.

The mitigation strategies discussed include a transition to low-carbon energy, reforestation, and carbon capture technologies.

Attributing disaster damage to climate change remains complex. Bouwer (2011) critically examine trends in economic losses and finds that much of the observed increase can be explained by rising exposure—such as more people and infrastructure in vulnerable areas—rather than a clear signal from climate change alone. This work underscores the difficulty of isolating climate-driven changes in risk from socioeconomic factors as a persistent challenge.

2.2 Modeling natural disasters and damage

Numerous efforts have been developed by climate scientists to develop accurate predictions of extreme weather events independently of their economic consequences. Since this paper focuses on economic impacts, we emphasize literature addressing this aspect of the research problem.

An important focus of past research has been the development and application of Integrated Assessment Models (IAMs). However, it is important to note that IAMs often lack key components. Most notably, they do not incorporate a geospatial representation of disasters, which is central to the present research. For this reason, we adopt a different modeling approach better suited to addressing this gap.

In his Nobel Prize lecture William Nordhaus argues that climate change is the ultimate challenge for economics as an academic discipline, and asserts that "*projecting impacts is the most difficult task and has the greatest uncertainties of all the processes associated with global warming*" (Nordhaus, 2019). He also provides a clear guide through past efforts undertaken by the discipline of economics to understand and model the economic impacts of climate change.

Nordhaus (2019) emphasizes the importance of models able to integrate different systems that are deeply integrated in the real world, but that are approached with a segmented perspective by different academic disciplines. In this sense, Nordhaus recognizes the relevant role of Integrated Assessment Models (IAMS) as the powerhouse tool for economic modeling of climate change.

Botzen et al. (2019) develop a valuable literature review about the state of the art. One of their findings is that empirical studies on the economic impacts of natural disasters rely heavily on various data sources, with the Emergency Events Database (EM-DAT) (Centre for Research on the Epidemiology of Disasters, 2024) being the most commonly used. Compiled by the Centre for Research on the Epidemiology of Disasters (CRED), EM-DAT provides extensive data on natural disasters, although it has limitations such as variable thresholds for event inclusion and potential inflation of damage estimates. According to the authors, other databases like NatCatSERVICE and Sigma, created by reinsurance companies, are less frequently used due to their limited public availability. Recent studies have shifted towards using geophysical or meteorological variables to define natural disasters. These physical indicators provide a more objective basis for assessing the economic impacts of disasters.

Kahn (2005) takes an innovative approach by modeling disasters frequency and impacts as a function of different economic, institutional, and development indicators. He explores the factors influencing the death toll from natural disasters across various nations. Using a comprehensive dataset on annual deaths from disasters in 57 countries from 1980 to 2002, Kahn (2005) tests several hypotheses regarding the impact of national income, geography, and institutional quality on disaster fatalities. The study reveals that while richer nations do not experience fewer natural disaster events than poorer nations, they do suffer significantly fewer deaths from these events. This suggests that economic development provides a form of implicit insurance against the adverse effects of natural disasters. Additionally, the paper finds that democracies and nations with higher quality institutions tend to have lower death tolls from natural disasters, highlighting the importance of governance and institutional frameworks in mitigating disaster impacts.

The analysis in Kahn (2005) utilizes both probit models and zero-inflated negative binomial regressions to examine the determinants of disaster fatalities. To identify the causal effect of geography on disaster fatalities, the study addresses potential confounding from national population density and institutional quality. By conditioning on democracy levels and income inequality, it isolates the specific role of geographic location. The analysis finds that geographic factors significantly influence death tolls. It also concludes that nations in Asia and the Americas experiencing higher fatalities relative to Africa. Furthermore, the paper underscores the critical role of institutional quality in insulating populations from the devastating effects of natural disasters.

López et al. (2015) offer a different perspective. They model the frequency of climate change related disasters as a function of country-level development indicators and global climate variables. They use a generalized linear model (GLM) to model the latent probability of a disaster with a set of country level covariates. Their approach does not consider the feedback dynamics included in IAMs models. Nevertheless, they provide a direct approach for linking climate variables, economic circumstances, and disasters.

Howard and Sterner (2017) perform a complete meta-analysis on the topic of global climate damage estimates to better understand the relationship between temperature changes and climate damages. They identify and address several biases present in previous meta-analyses, including duplicate estimates, omitted variables, measurement errors, overreliance on published estimates, dependent errors, and heteroskedasticity. They conclude that previous meta-analyses suffered from significant biases that flattened the relationship between temperature and climate damages. The authors find strong evidence that duplicate and omitted variable biases significantly impact the estimated temperature-damage relationship. Specifically, the magnitude of the bias depends on the treatment of speculative high-temperature damage estimates. When the authors replace the DICE-2013R damage function with an adjusted estimate, they find a three- to four-fold increase in the 2015 Social Cost of Carbon (SCC) relative to the DICE model, depending on the treatment of productivity. When catastrophic impacts are also factored in, the SCC increases by four- to five-fold. This suggests that previous estimates of climate damages and the SCC may have been significantly underestimated due to biases in the data and methodologies used.

In conclusion, the literature in this field has grown steadily over the past two decades, but challenges still remain. The integration of granular, geospatial approaches with broader macro perspectives remains limited. Despite the advantages of adopting a probabilistic perspective to better understand disasters and their associated damages, the application of Bayesian methods remains relatively uncommon. It is along these two fronts that we believe our contribution is most relevant.

3 Modeling Approach

3.1 General approach

We focus on *natural disasters* as a key channel of impact of the climate on the economy. We consider this to be a particularly relevant research problem as:

1. Climate researchers widely agree that disasters will likely become more frequent and more severe as the result of climate change.
2. Quantification of damages from discrete incidents of natural disaster is a tractable problem.
3. Specific actions related to disaster preparedness and prevention can be recommended to policy makers.

Our objective is to quantify countries' exposure risk to disaster events. In particular, we focus on what López et al. (2015) define as hydrometeorological disasters, which comprises storms, floods, and movements of wet masses. The same methodology can, however, be applied to climatological disasters, which include droughts and wildfires. Even when both disaster categories are deeply interconnected with climate dynamics, both require a separate treatment, as their causal relationships with precipitation differ significantly.

For the estimations we use a Bayesian modeling approach implemented through PyMC (Abril-Pla et al., 2023), a probabilistic programming language implemented in Python. This probabilistic approach offers us three crucial advantages:

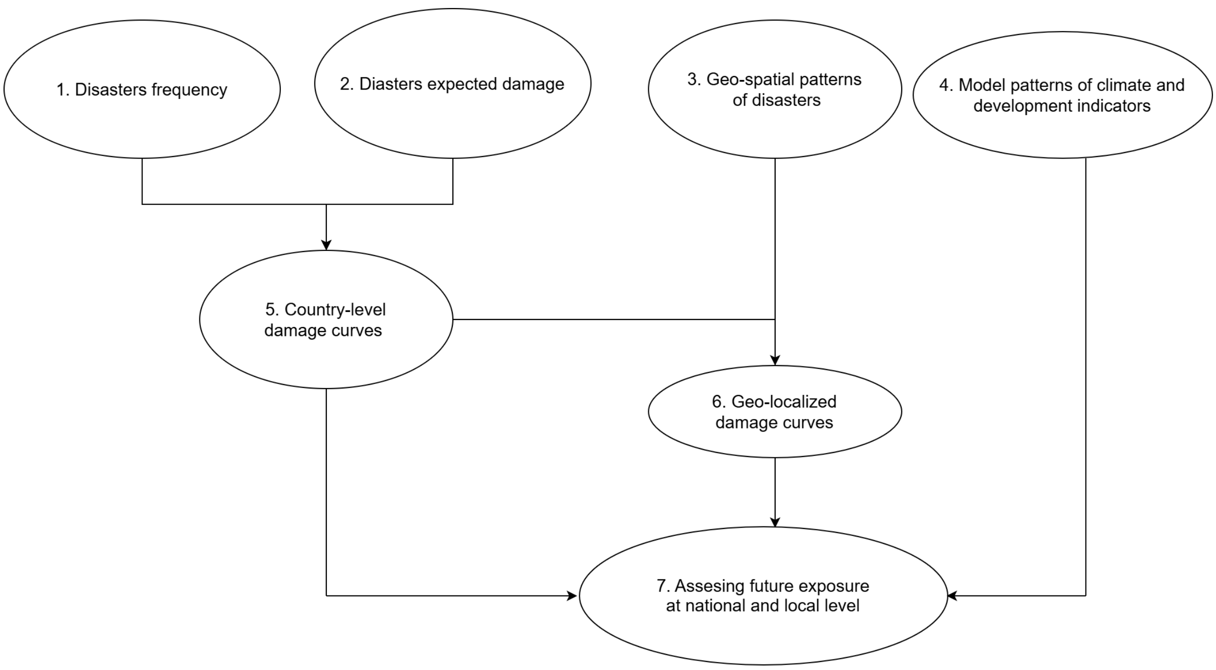
- First, the Bayesian framework facilitates the use of flexible Likelihood functions that more accurately represent the stochastic nature of disasters. Unlike traditional frequentist approaches—which often rely on asymptotic normality for inference or specific error-term assumptions (such as Gauss-Markov assumptions in OLS)—Bayesian modeling allows us to explicitly define non-normal generative processes. This is particularly relevant for disaster data, such as frequency and damage costs, which typically follow heavy-tailed or skewed distributions (e.g., Gumbel or Power Law) that are poorly captured by standard linear frameworks.

- Second, it allows us to estimate the entire probability distributions for events frequency and expected damages instead of just the expected values. This is extremely important, as this allows us to study the tails of the probability distributions where some of the most impactful events are located. By this approach we are able to quantify country exposure to different magnitudes of disasters.

- Third, it allows us to propagate uncertainty from one piece of the model to the other. This means that rather than treating intermediate estimates (such as disaster frequencies or economic vulnerabilities) as fixed inputs, we incorporate their full uncertainty into the final output. As a result, our estimates of exposure risk more accurately reflect the real-world ambiguity and variability inherent in climate-related data.

The modeling approach consists of seven components, illustrated in Figure 1. We present here a brief description of each section, however in further sections we will develop in detail each of the steps.

Figure 1: Modeling approach



1. Modeling disasters frequency:

First, we use global data on disasters, climate and development indicators to model disaster frequency and predict the probability distribution of the number of disasters each country will face per year.

2. Modeling expected disasters damage

We use the same global data on disasters, climate, and development indicators to model the probability distribution of the expected damages for each country over time.

3. Modeling geospatial patterns of disasters

Here we take a more granular approach to model the geospatial patterns of disasters inside countries. For this we use country maps and geolocalized information of disasters, rivers, and coastlines. We combine this information with global climate time series and development indicators to compute the probability of a disaster happening for each specific point on a country map. To model the geospatial patterns, we use a Gaussian Process, a non-parametric modeling technique with great power and flexibility.

To overcome some limitations of the geospatial information available in EM-DAT data set, we build an automatized query that uses a large language model to map some of the disaster-location information from text to geospatial coordinates. We consider this to be a relevant contribution to the current methodology in the field.

In contrast to the first two models, which are estimated simultaneously using data from all countries with available observations, the geospatial component is estimated using data from a specific region only, and predictions are generated for a single target country. This approach is necessary due to the substantial computational resources required to estimate the model at a global scale. To illustrate the methodology, we present results for two countries: Laos and Costa Rica.

4. Modeling patterns of climate and development indicators

In order to assess countries' future exposure, we model the time series patterns of climate and atmospheric indicators, as well as country-level development indicators. We use Bayesian Exponential Smoothing (ETS) models with additive damped trends, estimated via Kalman filtering, to capture level and trend dynamics in the data. We use this information to build future projections of key model variables.

5. Country-level damage curves

By combining the (1.) modelled disaster frequency and (2.) expected disasters damage, we can create country-level damage curves, that measure the expected damage the country will experience conditional on the expected number of disasters, and their probability distribution. These damage curve computations are expressed using the "return years" notion. Return years describe the average frequency at which a certain level of damage is expected to be met or exceeded. For example, a 100-RY event (or simply a "100-year" event) corresponds to a damage level that has a 1% chance of being surpassed in any given year, or in other words, an event of such magnitude that is expected to happen on average once every 100 years.

6. Geolocated damage curves

Combining these country level damage curves (5) and the geospatial projection of the probability of a disaster (3), we build geolocalized damage curves, which measure the exposure of every point on a map to disasters.

7. Assessing future exposure to climate change related disasters at national and local level

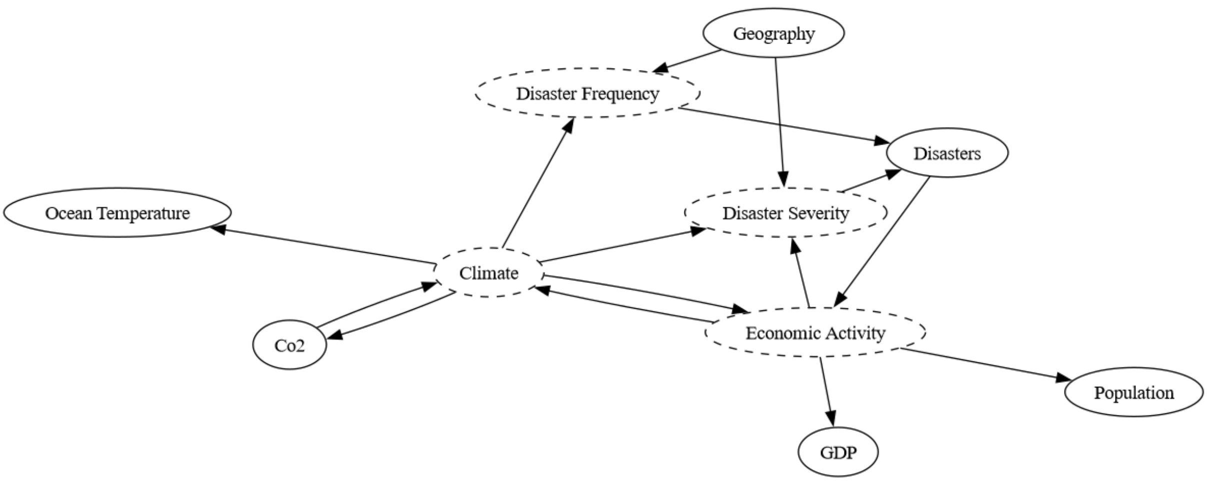
Finally, we combine all the previous outputs to build projections of event frequency, damages, geospatial patterns of disaster occurrence, and country-level damage curves, as well as geolocalized damage curves.

3.2 Underlying causal assumptions

To reason about our causal assumptions, we employ a causal model à la Pearl (2009). Figure 2 shows the causal assumptions underlying our approach. "Economic Activity" and "Climate", as complex, abstract systems, are unobserved processes. Instead, we observe proxies for these: GDP or population growth (in the case of the economy), or climatic indicators like atmospheric CO_2 or ocean temperatures (in the case of the climate). The climate, together with the location of a specific place, determine the frequency and intensity of disasters. Economic activity also contributes to disaster severity. This occurs when resources are being deployed towards adaptation and resilience (e.g. with dykes, levees, and floodwalls), or when overexploitation creates new risks (such as mudslides resulting from overdevelopment). More economic activity and higher GDP also imply more potential for damages, either from destruction of expensive infrastructure or from interruptions to economic activity. Of course, we don't actually observe the true frequency ($f_P(p)$) or severity ($f_D(d)$) variables; these we are left to infer by observing the disasters that do occur.

Figure 2: Causal Graph Between Economic Activity, Climate, and Disasters

Dotted circles represent unobserved variables. Arrows connect concepts causally, so that $A \to B$ implies that A causes B.



The causal line between economic activity and disaster severity has a more quotidian interpretation, related to our definition of "severity". That is, we adopt an economic definition, wherein the severity of a disaster is measured by the cost of the damage it inflicts. As such, there must first be things to damage as a necessary precondition. The more things that exist, the more things that can be destroyed, and the greater potential for disaster impact.

Importantly, Figure 2 is silent on the specific functional relationships between variables. Each arrow on this diagram represents an entire discipline of study as shown in the literature review. Our objective is not to model these causal linkages individually, but to model them jointly.

4 Methodology

4.1 Uncertainty quantification

Uncertainty quantification lies at the heart of our modeling approach. Disaster damage data typically exhibits positive skew and long right tails, where a small number of extreme events drive the majority of observed impacts. Consequently, traditional mean-based metrics fail to capture the reality of catastrophic risk. Instead, it is the tails of the distribution that demand attention. To address this, we employ generative probabilistic modeling (Gelman et al., 1995) as a framework for causal inference. This approach allows us to move beyond passive observation and perform counterfactual interventions using Judea Pearl's do-operator (Pearl, 2009). By simulating the full distribution of potential outcomes under the intervention $P(\text{Damage} \mid \text{do}(\text{Policy}))$, we can conduct rigorous scenario analysis that accounts for functional relationships, prior research, and the inherent uncertainty in the data.

While many generative frameworks exist, PyMC (Abril-Pla et al., 2023) focuses on Bayesian generative models. This means that not only do our models output probability distributions, we also specify probability distributions over model parameters. These parameters, jointly denoted θ, can then be updated according to Bayes' theorem:

$$p(\theta \mid X, \mathcal{M}) = \frac{p(X \mid \theta, \mathcal{M}) p(\theta \mid \mathcal{M})}{\int p(X \mid \theta, \mathcal{M}) p(\theta \mid \mathcal{M}) \mathrm{d}\theta} \tag{1}$$

Where X is observed data and $\mathcal{M}$ is a model parameterized by θ. The left-hand quantity, $p(\theta \mid X, \mathcal{M})$, is an updated belief about the possible parameter values that θ could take on, given that we've seen data X. This is called a *posterior*. The integral in the denominator of this expression is the sum of every possible parameter vector that could have given rise to the observed data, weighted by how likely we believe each parameter vector to be. This is infamously impossible to compute. Instead, the posterior can be approximated using sequential sampling schemes known as Markov Chain Monte Carlo (MCMC) algorithms. PyMC offers state-of-the-art MCMC algorithms, notably the No-U Turn Sampler (NUTS) (Hoffman et al., 2014).

The posterior is not the central object of interest in our research. Instead, we are interested in the range of possible disaster impacts that are implied by that posterior. This leads us to the *posterior predictive distribution*:

$$p(y \mid X, \mathcal{M}) = \int p(y \mid X, \theta, \mathcal{M}) p(\theta \mid X, \mathcal{M}) \mathrm{d}\theta \tag{2}$$

Where y is a new piece of data, not included in X. This expression marginalizes the uncertainty arising from the posterior, and gives a parameter-free probability distribution over possible outcomes, given the modeling framework and the input data. Armed with the posterior predictive distribution, we can answer important distributional questions. This will be extremely useful for building damages curves and damage curves predictions, as we explain next.

4.2 Modeling disaster frequency

As a first step, we model disaster frequency at country level. We use disaster data from the EM-DAT data set (Centre for Research on the Epidemiology of Disasters, 2024). We model the number of disasters a country experiences per year as a function of climate variables and country-level development indicators. We focus on one type of disasters: hydrometeorological (storms, floods, and movements of wet masses).

To model the number of disasters we use a Zero Inflated Binomial probability distribution, which allows us to better capture the fact that the generative process of disasters has a big mass concentration around zero, as multiple years do not report disasters for several countries. Consequently, we can say that the number of disasters occurring in country i on year t is distributed according to the Zero Inflated Binomial probability distribution:

$$Y_{i,t} \sim \text{ZeroInflatedBinomial}(\Psi, \mu, \alpha) \tag{3}$$

The Zero Inflated Binomial probability is completely defined by three parameters:

- **Inflation Parameter** Ψ: represents the probability of an observation being an excess zero. It models the likelihood that a zero count comes from the zero-generating process rather than the Binomial process.
- **Mean parameter** μ: This is the mean parameter of the Negative Binomial distribution. It represents the average number of events (or counts) you would expect to see if the observation is not an excess zero.
- **Variance parameter** α: This is the dispersion parameter of the Negative Binomial distribution. It controls the variance of the distribution. Smaller values of α lead to more over-dispersion (greater variance relative to the mean), while larger values make the distribution approach a Poisson distribution.

Each of these three parameters is modeled with a probability distribution. In this way, we have a nested structure in which the Zero Inflated Binomial distribution that accounts for events frequency is expressed as a function of other three probability distributions. It is typical when working with a GLM to make the expected value μ the center of attention in the model, by writing it as a transformed linear function of data. The present study is no exception. We model

$\mu = \mathbb{E}[Y_{i,t} \mid X, \theta]$, the expected count of disasters in country i during year t, as a function of observed covariates and parameters θ.

To model Ψ, the probability of observing zero disasters for a given country-year tuple, we use a Beta probability distribution. The Beta distribution is a family of continuous probability distributions defined on the interval $[0, 1]$. It is parameterized by two positive shape parameters, typically denoted by α and β, which we will denote with $\tilde{\alpha}$ and $\tilde{\beta}$, to differentiate them from parameters in the latent linear model of μ:

$$f(x|\tilde{\alpha}, \tilde{\beta}) = \frac{x^{\tilde{\alpha}-1} \cdot (1-x)^{\tilde{\beta}-1}}{B(\tilde{\alpha}, \tilde{\beta})} \tag{4}$$

Where $B(\tilde{\alpha}, \tilde{\beta})$ is a normalization function of the form:

$$B(\tilde{\alpha}, \tilde{\beta}) = \int_0^1 s^{\tilde{\alpha}-1}(1-s)^{\tilde{\beta}-1} \mathrm{d}s \tag{5}$$

To model the α parameter we use a Gamma probability distribution. The Gamma distribution is a two-parameter family of continuous probability distributions. It is widely used in statistics due to its flexibility in modeling positive, continuous data. The Gamma distribution is strictly positive and completely defined by two parameters:

- **Shape parameter** k: This parameter influences the shape of the distribution. It is also known as the "shape factor."
- **Scale parameter** θ: This parameter scales the distribution.

The full probability density function is given as:

$$f(x|k, \theta) = \frac{x^{k-1} e^{-x/\theta}}{\theta^k \Gamma(k)} \tag{6}$$

Where $\Gamma(k)$ corresponds to the Gamma function:

$$\Gamma(k) = \int_0^\infty t^{k-1} e^{-t} \mathrm{d}t \tag{7}$$

Now we move to the core of the model, which is given by the parameter μ, which we use to link a linear model to the expected disaster frequency. We assume the relationship between μ and the observed covariates X is linear in some latent space. A link function is chosen to transform the latent linear space to the space of observations, which is strictly positive. Thus, for every country i, in every year t, we compute $\mu_{i,t}$ using the functional form:

$$\mu_{i,t} = \text{softplus}(Z_{i,t}) \tag{8}$$

Where $Z_{i,t}$ is the latent "rate" at which we observe disasters, modeled as a linear function of data. The softplus function is a continuous approximation of the function $f(x) = \max(0, x)$[1]:

$$\text{softplus}(Z_{i,t}) = \log(1 + e^{Z_{i,t}}) \tag{9}$$

Where we define the latent rate Z as the sum of three components:

$$Z = \text{country\_fixed\_effect}_i + \text{spatial\_random\_effect}_i + \text{covariates\_effect}_{i,t} \tag{10}$$

These components have the following meanings:

- country_fixed_effect is a model intercept with a country hierarchy. This means that it is modeled using a Bayesian hierarchical framework, which serves as an intermediary between the extremes of no-pooling (independent OLS per country) and complete pooling (a single global intercept). This structure assumes that country-level intercepts are drawn from a common global distribution, allowing for partial pooling of information across units. For countries with sparse data or extreme outliers, the hierarchical prior induces shrinkage of the country-specific estimates toward the global mean. This "information sharing" significantly improves the reliability of estimates for small-sample countries, effectively reducing the noise and over-fitting.
- spatial_random_effect$_i$ accounts for spatial autocorrelation. To account for spatial autocorrelation, we incorporate a Besag-York-Mollié (BYM) specification (Besag et al., 1991). This approach decomposes the spatial_random_effect$_i$ into a mixture of two distinct components to distinguish between regional trends and localized anomalies:
  - A global latent spatial term $\tilde{\theta}$. Captures global, unstructured variation. If a country behaves very differently from its neighbors (e.g., due to policy, geography, or data quality), this term allows for that deviation.
  - A structured spatial term from an Intrinsic Conditional Autoregressive (ICAR) model ϕ. The ICAR model assumes that the value of a country's spatial random effect is similar to that of its neighbors. Mathematically, this is a Gaussian Markov Random Field (GMRF) with dependence defined through an adjacency matrix W. This structure introduces local smoothing: nearby countries' effects "pull" each other.

These two terms combined using a convex combination (weighted average) determined by ρ, a mixing coefficient:

$$\text{spatial\_random\_effect}_i = \sqrt{1-\rho} \cdot \tilde{\theta}_i + \sqrt{\frac{\rho}{s}}\phi_i \tag{11}$$

[1]This max function is called the ReLU function in the deep learning literature. In the GLM literature, the canonical link function for a Zero Inflated Binomial is the exponential function. We use softplus for its superior numerical properties. In particular, we did not want exponential blowup in the mean with respect to the input data.

With s a scaling factor that ensures that the variance of the ICAR component is appropriately normalized.

For the covariates_effect$_{i,t}$ we introduce the time dimension t, and we use the following covariates:

- ln_population_density_standardized$_{i,t}$: log of population density for country i, period t.
- ln_gdp_pc_standardized$_{i,t}$: log of GDP per capita for country i, period t.
- population_standardized$_{i,t}$: Population in millions for country i, period t squared.
- precip_deviation_standardized$_{i,t}$: Deviation of precipitation on country i on period t from its trend.
- co2_standardized$_t$: Worldwide atmospheric CO2 concentration in parts per million at period t.

To control for unobserved heterogeneity on the panel structure, we apply the Mundlak adjustment by introducing group-level (country-level) random slopes for each covariate, which allows the model to account for correlation between covariates and latent group-specific effects. This way, the covariates functional form has the structure:

$$\text{covariates\_effect}_{i,t} = \sum_{j=0}^{n} X_{i,t,j} \cdot \beta_j + \sum_{j=0}^{n} \lambda_{i,j} \cdot \beta_j^{\text{latent}} \tag{12}$$

Where:

- i corresponds to country index, t year index, and j covariate index.
- $\lambda_{i,j}$ is a country-specific deviation (random slope) for covariate.

4.3 Modeling disaster damages

For disaster damage, we use the logarithm of disaster damage. We model this variable using a Hurdle LogNormal probability distribution. The Hurdle LogNormal distribution is a type of hurdle model used to handle zero-inflated continuous data. Hurdle models are particularly useful when dealing with datasets that have an excess of zero values, in addition to continuous positive values. The Hurdle LogNormal model specifically combines a binary process that models the zeros and a truncated LogNormal process that models the positive continuous values.

The overall likelihood for the Hurdle LogNormal model is the product of the likelihoods from the binary component and the truncated LogNormal component:

- p if $y_i = 0$

- $(1-p) \cdot \text{LogNormal}(y_i|\mu, \sigma)$ if $y_i \neq 0$

Here, the LogNormal component is completely defined by the parameters μ and σ. To model sigma, we use a Half Normal distribution, which ensures strictly positive terms. While for μ, we use exactly the same linear structure we used for frequency μ on equation 10.

4.4 Modeling geospatial patterns of disasters

As mentioned previously, the geospatial component is estimated using data from a specific region only, and predictions are generated only for a single target country at a time. This approach is necessary due to the substantial computational resources required to estimate the model at a global scale. To illustrate the methodology, we present results for two countries: Laos and Costa Rica.

In the case of the probability of the event model, the objective is to model the probability of one event happening at any point of a country's grid. To achieve this, we create a set of synthetic non-disaster points, and we train the model to identify the probability of a point being a real disaster or a synthetic non-disaster point.

We estimate two different model versions. The first uses only the coordinates (latitude and longitude) of the grid and disaster points to capture geographical patterns of disaster occurrence. It captures these patterns by applying a Hilbert Space Gaussian Process (HSGP).

The second one incorporates, on top of the HSGP model, the distance to rivers and coastlines variables, plus a set of country-level covariates and climate-related time series.

In both cases, we estimate the probability of occurrence of an event by using a logit regression estimated in a Bayesian framework with PyMC (Abril-Pla et al., 2023). Therefore, we model the probability of an event happening at location g in country i at time t. This way, the general specification corresponds to the general form:

$$P(Y = 1 \mid X)_{i,g,t} = \sigma \left(\text{country\_fix\_effect}_i + \text{HSGP}_{i,g} + X_{i,g,t} \cdot \beta_{i,g,t} \right) \tag{13}$$

Where:

- $\sigma(Z)$ corresponds to the logistic function $\frac{1}{1+e^{-z}}$.
- country_fix_effect$_i$ represents the country fix effect for country i.
- HSGP$_{i,g}$ is the Hilbert Space Gaussian Process component that will be explained next.
- $X_{i,t} \cdot \beta_{i,t}$ corresponds to a set of covariates and their parameters, where we include three types of covariates: geospatial, climate time series, and country-level macro variables.

Hilbert Space Gaussian Process

The **Hilbert Space Gaussian Process (HSGP)** is an approximation method used to represent a Gaussian Process (GP) in a computationally efficient manner. Traditional GPs require inverting an $N \times N$ covariance matrix, which becomes computationally expensive for large datasets. HSGP approximates the GP using a finite basis function expansion, reducing the complexity.

The function $\phi_k(x)$ and the corresponding spectral densities λ_k to approximate the GP prior:

$$f(x) \approx \sum_{k=1}^{m} \beta_k \sqrt{\lambda_k} \phi_k(x) \tag{14}$$

where:

- $\beta_k \sim \mathcal{N}(0, 1)$ are the basis coefficients.
- λ_k are the eigenvalues derived from the covariance function.
- $\phi_k(x)$ are the basis functions, which capture spatial variations.

Our model applies an HSGP prior to analyze the probability of a geospatial location given by (lat, long) of experiencing a climate-related disaster. The key components are:

1. **Spatial Covariance Function:**

$$k(x, x') = \eta^2 K(x, x'; \ell) \tag{15}$$

   where η is a scale parameter and ℓ is the length scale, drawn from a lognormal prior.

2. **Hilbert Space Representation:** The GP is approximated using a Matérn-5/2 kernel with parameters (m_0, m_1) controlling the number of basis functions. We use 35 as a value for both parameters. The basis function matrix Φ and spectral density $\sqrt{\lambda}$ define the GP prior:

$$\Phi, \sqrt{\lambda} = \text{gp.prior}_{\text{linearized}}(X) \tag{16}$$

3. **Deterministic Component:**

$$\text{HSGP}_{\text{component}} = \Phi \cdot (\beta \odot \sqrt{\lambda}) \tag{17}$$

   where $\beta \sim \mathcal{N}(0, 1)$ are the basis coefficients.

4. **Bernoulli Likelihood:** The probability of a disaster occurring at a given location is modeled as a Bernoulli process with a logit link:

$$p = \text{HSGP}_{\text{component}}, \quad y \sim \text{Bernoulli}(\sigma(p)) \tag{18}$$

   where $\sigma(p)$ is the logistic function mapping to probabilities.

Given this specification, the HSGP model leverages geospatial coordinates (latitude and longitude) as inputs, allowing it to learn spatial dependencies in disaster occurrences. By reducing computational complexity, HSGP enables inference on large datasets where standard Gaussian process methods would be infeasible. The prior structure ensures smooth spatial predictions, capturing meaningful regional variations in disaster risk. This whole setup allows the model to capture the likelihood of climate-related events in different regions while maintaining scalability.

Full model specification

In this second step, we estimate the full model specification:

$$P(Y = 1 \mid X)_{i,g,t} = \sigma\left(\text{country\_fix\_effect}_i + \text{HSGP}_{i,g} + X_{i,g,t} \cdot \beta_{i,g,t}\right) \tag{19}$$

Here, in addition to the geospatial pattern detected by the HSGP, we obtain for every point in the grid of the region analyzed and for every disaster point the variables corresponding to 3 sets of covariates: climate change series, country-level specific variables, and distances to rivers and coastlines. All the covariates we introduce here are standardized to simplify the scale manipulation and the prior setting in the estimation. The standardization is done based on the whole model data set's mean and standard deviation. All the priors of the covariates parameters are defined as normal distributions with mean zero and standard deviation one.

The full set of covariates is:

<u>Climate change series</u>:

- $\text{co2\_standardized}_t$: Worldwide atmospheric CO2 concentration in parts per million at period t.
- $\text{dev\_ocean\_temp\_standardized}_t$: Deviation of the mean world's ocean temperature from its trend on period t.
- $\text{precip\_deviation\_standardized}_{i,t}$: Deviation of precipitation on country i on period t from its trend.

<u>Country-level specific variables</u>:

- $\text{ln\_population\_density\_standardized}_{i,t}$: log of population density for country i, period t.
- $\text{ln\_gdp\_pc\_standardized}_{i,t}$: log of GDP per capita for country i, period t.
- $\text{square\_ln\_gdp\_p\_standardized}_{i,t}$: log of GDP per capita for country i, period t squared.
- $\text{population\_standardized}_{i,t}$: Population in millions for country i, period t squared.

<u>Distances to rivers and coastlines</u>:

- For each point of the map, we compute the distance with the closest river, and we introduce its log as feature: log_distance_to_river_standardized$_g$.
- We also compute the distance to the closest coastline, and we include its log as feature: log_distance_to_coastline_standardized$_g$.

4.5 Modeling patterns of climate and development indicators

With the objective of being able to make predictions, we use a State Space modeling framework to predict trends in key inputs of the model. In particular, we use the implementation of State Space modeling via PyMC (Grabowski, 2023). In particular we project for a specific country of analysis: real GDP, population, deviation from trend precipitation. Additionally we project the global variables: atmospheric CO_2 accumulation, and ocean temperature deviations from trend.

For this, we employ a Bayesian dynamic linear model based on the PyMC state-space module, using a Bayesian Exponential Time Smoothing (ETS) (Hyndman et al., 2008) approach. This approach enables probabilistic inference and forecasting under a fully Bayesian framework. We model the multivariate time series using an additive trend exponential smoothing state-space model with the following configuration: local level (A), damped local trend (Ad), and no seasonal component (N). The model specification includes measurement error, a dense innovation covariance structure for the state shocks, and a stationary initialization with dampening. This setup provides flexibility to accommodate a wide range of time-varying behavior, including long-run stochastic trends, persistent shocks, and changing volatilities across the multiple indicators.

Formally, let $y_t \in \mathbb{R}^n$ denote the observed vector of the n variables at time t, which evolves according to a linear Gaussian state-space system:

$$y_t = Z\alpha_t + \epsilon_t, \quad \epsilon \sim \mathcal{N}(0, H) \tag{20}$$

$$\alpha_t = T\alpha_{t-1} + R\eta_t, \quad \eta_t \sim \mathcal{N}(0, Q) \tag{21}$$

where α is the latent state vector (e.g., level and trend components), H is the measurement error covariance matrix, and Q is the innovation covariance matrix for the state shocks. These are learned from the data via a hierarchical prior specification, as described below.

Prior Elicitation with Maximum Entropy Transformations

To reflect informed but flexible beliefs about key smoothing and structural parameters, we employ maximum entropy–transformed priors (pz.maxent) for most hyperparameters. This allows us to specify constrained, information-efficient priors that respect known bounds or regularities (e.g., positivity, bounded damping).

Multivariate Covariance via LKJ Prior

We place an LKJ prior on the Cholesky decomposition of the state innovation covariance matrix Q, allowing for correlated innovations across state dimensions (e.g., co-movements between economic and climate variables). Specifically, we decompose $Q = LL^T$, where L is the lower triangular Cholesky factor, and the standard deviations and correlations are separately inferred. We set the concentration parameter $\eta = 4$, favoring weakly correlated structures but allowing for flexibility in learning strong correlations if supported by data. The standard deviation terms are assigned Gamma priors constrained to lie within plausible bounds.

Forecasting

Posterior Inference and Inverse Transformation: The model is then fit to data using Hamiltonian Monte Carlo (HMC) sampling via the nutpie backend (Langer and Developers, 2023) with jax acceleration (Bradbury et al., 2018). The forecasts and trajectories generated by the model are returned in standardized space; hence, we apply an inverse exponential transformation composed with a StandardScaler inverse to convert model outputs back into the original data units.

4.6 Country-level damage curves

Once we computed the probability distributions of event frequency, and the probability distribution of damages from points 4.2 and 4.3, the expected damages can be easily computed as a product of both:

$$P(\text{expected\_damage})_{i,t} = P(\text{frequency})_{i,t} \times P(\text{damages})_{i,t} \tag{22}$$

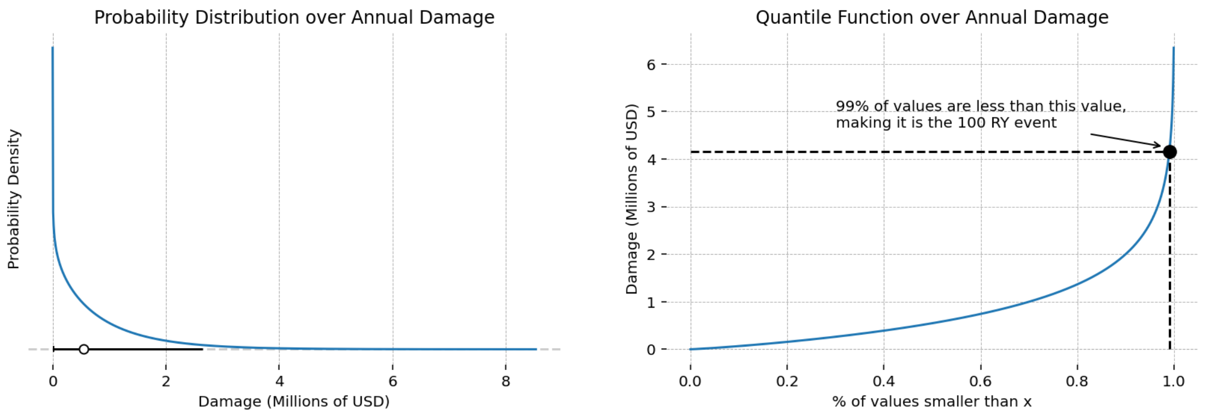
One advantage of this approach, and the Bayesian computations method we use, is that it allows us to propagate and quantify uncertainty from the initial two models to the expected damages.

Now that we have the expected damages, we resort to the notion of **return years** (RY) to represent the results in a much more intuitive way. A 100-RY disaster is one the associated damage of which is expected to be equaled or exceeded once every one-hundred years. Our primary object of study is, thus, a damage curve expressed in return years. The left panel of Figure 3 shows an example distribution over damages for a given year. The x-axis shows possible damage values, while the height is the probability density at that damage. Based on this curve, we can create a quantile function, which maps probabilities (on the x-axis) to the damage value which is larger than that percentage of values. The 100-RY damage value is larger than 99% of all damage values produced by this distribution, and is marked.

Several features of Figure 3 are worth pointing out. The curve has a lot of mass around zero, followed by a long tail. Large disasters are, after all, relatively rare, with many years seeing none at all. As a result, the mean of the distribution (marked with an open circle on the left plot) is not at all interesting from a policy perspective. Instead, we are interested in quantifying the density

Figure 3: Example Damage Curve for a Single Year and Disaster Type

The left curve shows all possible damage outputs for a category of disaster (e.g. flood, earthquake, drought, etc.). The x-axis shows the possible dollar-denominated damages, while the y-axis shows the probability density associated with that damage value (higher = more probable). The black bar shows the largest interval containing 95% of all possible outcomes (called the 95% HDI). The right curve shows the quantile of the distribution on the left. The 100-RY disaster, associated with the 99% quantile, is marked.



in the *tail*. Note that the 100-RY disaster, with a value of 4.1, is already a very rare event. But we can also contemplate even rarer events. In this example, a 500-RY event would have a value of 5.68. As the tail of the distribution becomes fatter, the difference between sizes of these rare events becomes larger.

Of primary interest to us is how climate change can potentially shift these curves, and how they vary according to geospatial patterns.

Climate researchers expect disasters to become both *more severe* and *more frequent* as the climate changes. We reason about these possibilities separately, by decomposing the problem in two parts.

As explained, the distribution shown in Figure 3 is computed by using two probability distributions: the probability distribution of seeing events, and the probability distribution of damages in case an event happens. Formally, we denote $D : \Omega \to \mathbb{R}^+$ as a random variable representing annual disaster damage, *conditional on a disaster occurring*. In addition, denote $P : \Omega \to [0, 1]$ as the probability of a disaster occurring. Their joint probability distribution is defined as:

$$f_{P,D}(p, d) = f_{D|P}(d|p) f_P(p) \tag{23}$$

This form allows for a shift in either the *frequency* of disasters, represented by $f_P(p)$, or the *severity* of disasters, represented by $f_{D|P}(d|p)$. To complete the model, we introduce a Bernoulli distributed random variable $I : \Omega \to \{0, 1\}$ with probability mass function:

$$f_{I|p}(i|p) = \begin{cases} p & \text{if } i = 1 \\ 1 - p & \text{if } i = 0 \end{cases} \tag{24}$$

The full model is thus:

$$f_{D,I,P}(d,i,p) = f_{D|I,P}(d|i,p)f_{I|P}(i|p)f(p) \tag{25}$$

The damage curve we observe in the data is the marginal distribution over damages, obtained by integrating over both P and I:

$$\begin{align} f_D(d) &= \int_0^1 \sum_{i\in\{0,1\}} f_{D,I,P}(d,i,p)\mathrm{d}p \tag{26}\\ &= \int_0^1 f_{D,I,P}(d,i=0,p)\mathrm{d}p + \int_0^1 f_{D,I,P}(d,i=1,p)\mathrm{d}p \tag{27}\end{align}$$

When $i=0$, $d=0$ by definition, so the expression simplifies significantly:

$$\begin{align} \mathbb{P}(D=0) &= \int_0^1 f_{I|P}(i=0|p)f(p)\mathrm{d}p \tag{28}\\ &= \int_0^1 (1-p)f(p)\mathrm{d}p \tag{29}\\ &= \int_0^1 f(p)\mathrm{d}p - \int_0^1 pf(p)\mathrm{d}p \tag{30}\end{align}$$

The first term here is 1, because $f(p)$ is a probability distribution. The second term is exactly the expected value of P. Defining $\mathbb{E}[P] = \mu = \int_0^1 pf(p)\mathrm{d}p$, the point mass on zero is simply:

$$\mathbb{P}(D=0) = 1-\mu \tag{31}$$

On the other hand, when $i=1$ then $d>0$ and:

$$\begin{align} f_D(d) &= \int_0^1 f_{D,I,P}(d,i,p)\mathrm{d}p \tag{32}\\ &= \int_0^1 f_{D|I,P}(d|i=1,p)f_{I|P}(i=1|p)f(p)\mathrm{d}p \tag{33}\\ &= \int_0^1 f_{D|I,P}(d|i=1,p)(p)f(p)\mathrm{d}p \tag{34}\end{align}$$

Assuming D and P are conditionally independent given $i=1$, this expression becomes:

$$f_D(d) = f_{D|I}(d|i=1)\int_0^1 pf_P(p)\mathrm{d}p \tag{35}$$

Once again the expected value of P appears, allowing the following simplification for the marginal distribution over disasters:

$$f_D(d) = \begin{cases} \mu \cdot f_{D|I}(d|i=1) & \text{if } d > 0 \\ 1 - \mu & \text{if } d = 0 \end{cases} \tag{36}$$

This is a zero-inflated mixture model with a point-mass on zero representing the years with no disaster. Written this way, the two proposed mechanisms to increase disasters are clear. When disasters become more frequent, it implies that μ becomes larger. This increases the multiplier on the damage term $f_{D,I}(d, i = 1)$, and shrinks the probability of not having a disaster. On the other hand, when disasters become more severe, it only scales up the damage term, while leaving the probability of not seeing a disaster unchanged. In either case, the expected damages from disasters increases. Starting from the definition of the expected value:

$$\mathbb{E}[D] = \int_0^\infty d f_D(d) \mathrm{d}d \tag{37}$$

$$= 0 \cdot (1 - \mu) + \mu \int_0^\infty d f_{D|I}(d|i = 1) \mathrm{d}d \tag{38}$$

$$= \mu \mathbb{E}[D|I = 1] \tag{39}$$

Again it is clear when μ becomes larger, the expected damages increase. Furthermore, $\mathbb{E}[D|I = 1]$ is the average severity of disasters. It is equally clear that when this increases, the $\mathbb{E}[D]$ also increases.

This formalism provides us a framework for thinking about the channels through which climate change can cause increased disaster risk. It is silent, however, on how to actually model the quantities of interest. For starters, D and P are not directly observed. We can only observe the disasters (storms, floods, etc) that actually happen, and try to infer something about their frequency over time. This frequency, however, is driven by many factors, including the climate, local geography where disasters are observed, as well as human activity, including economic activity. In addition, what we observe are imperfect proxies for the true quantities of interest, introducing additional layers of uncertainty. Here is where the causal assumptions presented in Figure 2 become highly relevant, as they provide a causal framework that justifies the statistical models we ultimately employ to estimate $f_P(p)$ and $f_D(d)$.

4.7 Geolocated damage curves

Once we have the country-level expected damages, we disaggregate these estimates spatially by leveraging our previously estimated probability of disaster occurrence at each geolocated grid point g. This allows us to construct geolocated damage curves, capturing how expected damages vary across space within each country. The key idea is to assign a share of national expected damages to each location, proportional to its relative disaster probability. This results in a spatially disaggregated risk map with expected damage curves for each grid point.

$$P(\text{expected\_damage})_{i,t,g} = \frac{P(Y=1|X)_{i,g,t}}{\sum_g P(Y=1|X)_{i,g,t}} \times P(\text{expected\_damage})_{i,t} \tag{40}$$

This approach maintains internal consistency with the national-level expected damage figures. It distributes damages in proportion to the local probability of experiencing a disaster, as computed using our HSGP-based spatial event model. The denominator ensures the probabilities sum to one within each country-year, making the spatial distribution interpretable as a relative risk-weighted allocation. Importantly, this method inherits the full uncertainty structure of the Bayesian event and damage models, allowing us to construct local credible intervals around the expected damages at each location.

Once the expected damages are mapped at the local level, we proceed in a similar manner as before to construct geolocated damage curves. That is, for each location g, we generate the probability distribution over damage levels using the joint Bayesian posterior of both event probabilities and damage given an event. This yields a distribution $f_{D,g}(d)$ for each point g from which we derive return-year quantiles (e.g., 100-RY or 500-RY events) for local areas.

Geolocated damage curves add critical granularity to the policy relevance of our framework. Different locations within a country face widely varying exposure and vulnerability profiles. For instance, coastal regions may face frequent flooding, while inland agricultural zones may suffer from drought. Our framework accounts for these spatial patterns by conditioning on local predictors such as distance to rivers, elevation, and population exposure, alongside the HSGP structure that learns spatial dependencies directly from the data.

Finally, a similar causal framework used to interpret country-level changes under climate scenarios applies locally. Increases in $P(Y=1|X)_{i,g,t}$ and $P(\text{events})_{i,t}$ derive in more frequent local disasters (e.g., more floods in coastal towns), while changes in $P(\text{damages})_{i,t}$ imply higher severity conditional on a disaster (e.g., stronger typhoons hitting a specific region). Spatially resolved return-year curves provide the basis for identifying which areas are most vulnerable, how risks change under counterfactual scenarios, and where targeted adaptation investments may be most needed.

5 Data

5.1 Data sets summary

To process the data for the project, we built an open source automatized pipeline that downloads and processes the data. The pipeline can be accessed in the project's repository **link**. We want to make this data pipeline publicly available as an exercise of transparency, to facilitate replication and because we are engaged promoters of open-source software.

The data sets used to build the Probabilistic GeoRisk model can be classified into three different categories:

1. Country-level information

Following the approach of López et al. (2015), we use the World Bank World Development Indicators (The World Bank, 2024*b*), which contain a panel of yearly information for almost all countries in the world. From this data set, we extract each country's GDP per capita, population density, and total population.

To obtain each country's deviation from its trend precipitation, we use the GPCC world precipitation data set (Schneider, 2022), which contains geospatial information all around the world. The data set provides the monthly total precipitation on a regular grid with a spatial resolution of 0.25° x 0.25°, 0.5° x 0.5°, 1.0° x 1.0°, and 2.5° x 2.5° latitude by longitude. Using each country's shapefile, we transform this geospatial information into a country-level time series from which we calculate the trend and the deviation with respect to that trend.

We also use the EM-DAT data set (Centre for Research on the Epidemiology of Disasters, 2024) to obtain the total number of disasters per country and the economic damages they caused. As found by Botzen et al. (2019), other data sets like NatCatSERVICE and Sigma provide detailed information about disasters, especially in terms of disaster geolocation. Nevertheless, we decided to use the EM-DAT database given its public availability, and its status as the academic standard.

When working with EM-DAT disasters register, it is important to take into account possible reporting biases, as disasters with less severe impacts may not have been registered. Additionally, it is possible that as the reporting capacities of countries increase, the rate of registration of these minor events changes. To address this, we include some initial filters by considering only disasters after 1980 that affected more than 1,000 people according to the reported impact. We also limit the analysis to hydrometeorological disasters, which comprise storms, floods, and movements of wet masses. By applying these filters, the number of disaster observations is reduced from 26,664 to 5,692.

2. Global-level climate time series

At the world time series level, we use NOAA's (National Oceanic and Atmospheric Administration) atmospheric CO2 concentration data set (NOAA, 2024), which provides the worldwide levels of CO2 atmospheric concentration in Gigatons. We also obtain the world's mean ocean temperature and its deviation from trend from the NOAA's Global Surface Temperature Dataset (Zhang et al., 2019).

3. Geospatial data sets

In terms of geospatial information, we use the shapefiles from the World Bank Official Boundaries (The World Bank, 2024*a*), which contain the official maps of the territory and administrative divisions for each country. We also use the world river data set from HydroSHEDS (Lehner and Grill, 2013), which contains the maps of all the rivers around the world. Additionally we take advantage of the geospatial coordinates for disasters contained in the EM-DAT data set (Centre for Research on the Epidemiology of Disasters, 2024), which provide the exact geographical zone affected by each event.

In terms of geospatial information the EM-DAT data set has an important limitation, as most of the disasters are missing the coordinates data. Nevertheless, practically all of them have the exact name or names of the place where the disaster took place. To overcome this limitation, we developed an automatic query for an artificial intelligence model which transformed the location into exact coordinates. Details about this procedure are given in the current section.

Table 1: Summary of Data Sources Used in our Modeling Pipeline

Table provides a title and organizational source for each dataset. See the main text for detailed citations associated with each dataset.

Title	Source	Summary
EMDAT International Disaster Database	CRED	Historical database recording disaster occurrences and impacts worldwide.
NOAA Atmospheric CO_2 Dataset	NOAA	Detailed records of atmospheric carbon dioxide levels collected over time for climate research.
NOAA Ocean Surface Temperature Dataset	NOAA	Historical measurements of ocean surface temperatures.
Global Precipitation Climatology Centre Dataset	GPCC	Historical records of precipitation at various levels of geographic resolution.
HydroSHEDS Global Hydrography Dataset	WWF	High-resolution geospatial data detailing river networks, along with flow data.
World Development Indicators	World Bank	Economic, social, and environmental indicators
World Bank Shapefiles of National Boundaries	World Bank	Maps of country and administrative borders.

5.2 Event frequency data

For the event frequency data, we transform the EM-DAT dataset to obtain a count of the total number of disasters per country per year. To do this, we assume that for countries with no recorded disasters in a given year, the count is zero. We then merge this dataset with the World Bank Development Indicators (The World Bank, 2024*b*), the CO_2 data from NOAA (2024), ocean temperature data from Zhang et al. (2019), and the global precipitation dataset from Schneider (2022).

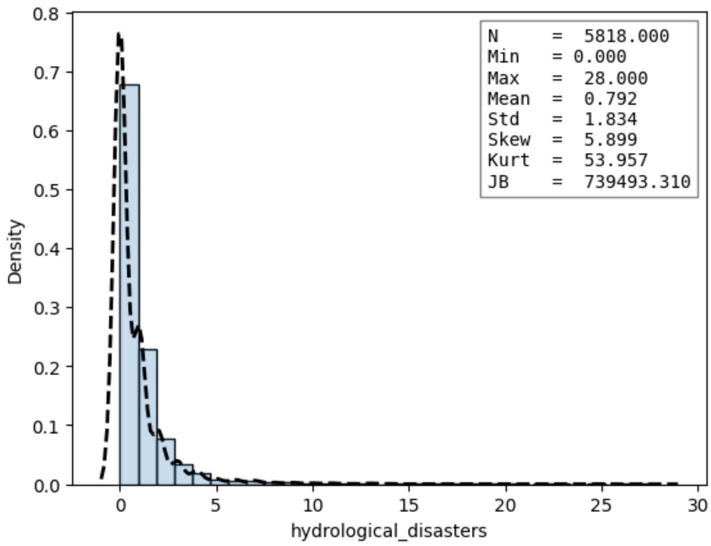
As previously mentioned, the precipitation data is provided at a geospatial level. Therefore, we use the World Bank shapefiles to assign each point to its corresponding country. We then compute the average annual precipitation per country, along with its deviation from the long-term trend. For the ocean temperature data, we similarly compute deviations from the historical mean. Table 2 presents summary statistics for the disaster event counts.

Table 2: Summary Statistics of Hydrological Disasters

This table presents the summary statistics for hydrological disaster data, including count, mean, standard deviation, percentiles, and measures of kurtosis and skewness.

Statistic	Value
Count	5818.00
Mean	0.79
Standard Deviation	1.83
Minimum	0.00
25th Percentile	0.00
Median	0.00
75th Percentile	1.00
Maximum	28.00
Kurtosis	54.00
Skewness	5.90

Figure 4: Kernel density for yearly count of hydrometeorological disasters by country



In Figure 5, we show the historical mean across all countries for the number of hydrometeorological events reported per year. A clear upward trend is observed. This may be partly due to reporting bias: as countries improve their institutional capacity, the number of recorded disasters tends to increase. However, it is important to note that we have already applied a filter to mitigate this bias by including only disasters that affected more than 1,000 people and occurred after 1980.

It is important to highlight that the GPCC precipitation dataset is only available up to 2020. To address this limitation, we estimate the model using data from 1980 to 2020. For predictions in more recent years, we use actual observed data for all covariates except precipitation, for which we rely on forecasts generated by a time series model.

Figure 5: Yearly mean across countries of the number of hydrometeorological disasters

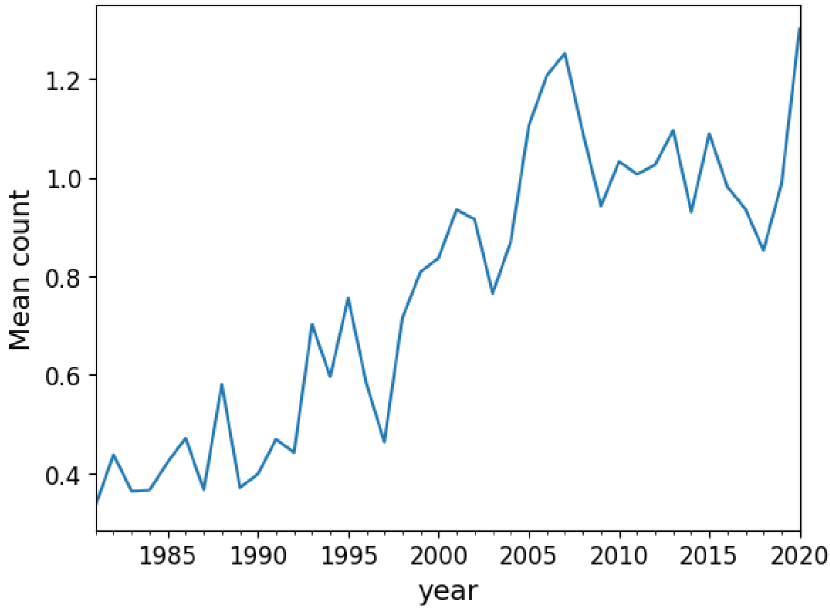


Table 3 and Figure 6 present a general summary of the data used as covariates in the model.

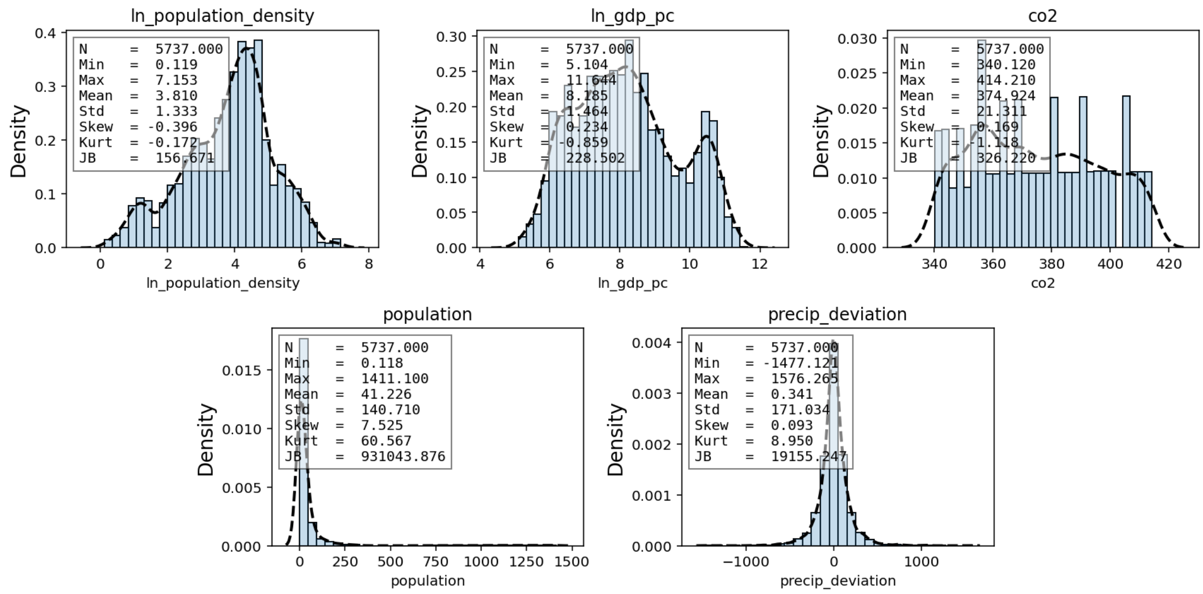
Table 3: Summary Statistics of event model covariates

Statistic	ln_pop_dens	ln_gdp_pc	co2	population	precip_dev
Count	5737.00	5737.00	5737.00	5737.00	5737.00
Mean	3.81	8.18	374.92	41.23	0.34
Std Dev	1.33	1.46	21.31	140.71	171.03
Min	0.12	5.10	340.12	0.12	-1477.12
25%	2.94	7.05	356.54	3.69	-69.37
Median	4.01	8.09	373.45	9.45	-1.17
75%	4.68	9.20	391.85	28.34	69.37
Max	7.15	11.64	414.21	1411.10	1576.26
Kurtosis	-0.17	-0.86	-1.12	60.62	8.96
Skewness	-0.40	0.23	0.17	7.53	0.09

5.3 Disaster damage data

Table 4 presents the summary for the damages data used for the damages model. One important limitation is that the number of disasters on the EM-DAT data set for which there is a reported damage value is considerably lower. As a consequence, the number of observations for the damage model is less than half of those in the event one: 2,300 vs. 5,692.

Figure 6: Kernel density for covariates included in the event frequency model



EM-DAT data set provides the damage values adjusted to 2011 USD, we adjust them to 2025 using the consumer price index time series from the U.S. Bureau of Statistics (U.S. Bureau of Labor Statistics, 2024).

Table 4: Summary Statistics of Total Damage Adjusted

This table presents the summary statistics for total damage adjusted in 2025 USD millions.

Statistic	Total_Damage_Adjusted
Count	2300.00
Mean	1721.12
Standard Deviation	8233.07
Minimum	0.00
25th Percentile	25.32
Median	172.87
75th Percentile	848.19
Maximum	261939.92
Kurtosis	507.95
Skewness	19.01

As seen in Figure 7, the damages data is characterized by extreme values reported in the biggest global economies like the United States, China and Japan. One clear example of this is the damages reported for hurricane Katrina in 2005, which have a 2025 adjusted value of 262 USD billions.

In contrast with the number of events registered, the damages of hydrometeorological disasters do not show a clear upward trend, as seen in Figure 8. Data reveals a peak of damages reported in 2005, a year where a set of highly destructive events were reported worldwide, including hurricane Katrina.

Figure 7: Kernel density for yearly damages of hydrometeorological disasters

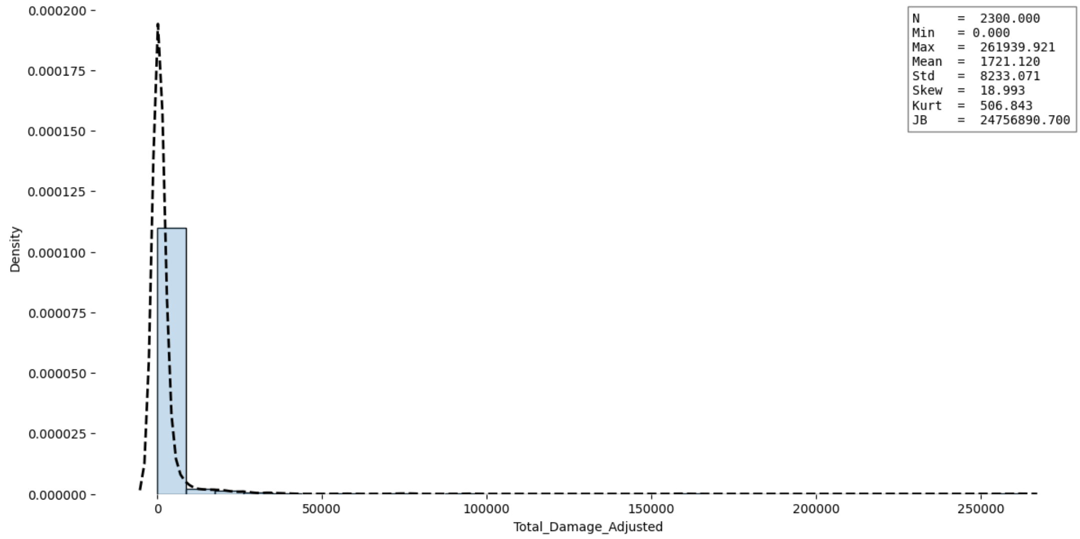


Figure 8: Historical trend of yearly damages of hydrometeorological disasters

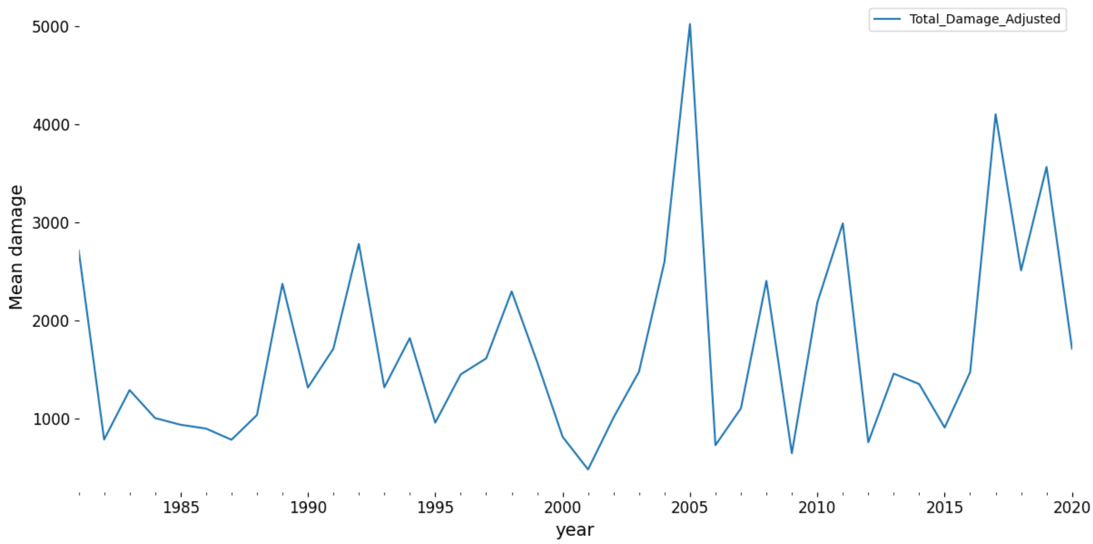


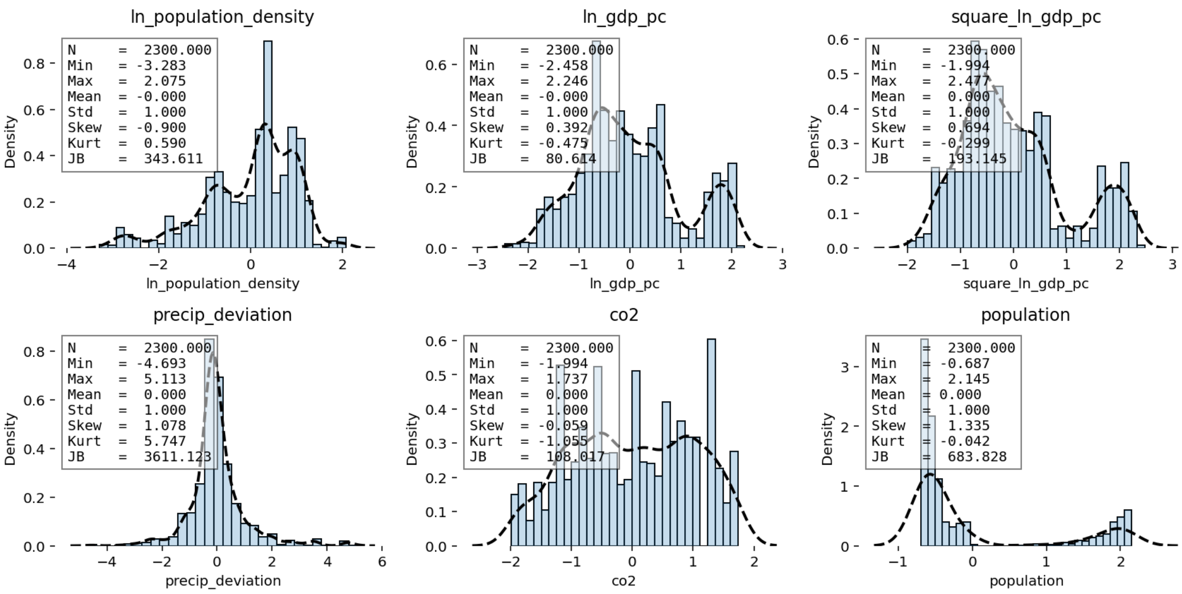
Table 5 presents the descriptive statistics for the features used in the damage model. In this case, to reduce the computing capacity required to estimate and sample the model, we standardize the features by subtracting the mean and dividing by the standard deviation.

Table 5: Summary Statistics of damage model covariates

All features have been standardized.

Statistic	ln_pop_dens	ln_gdp_pc	sq_ln_gdp	precip_dev	co2	population
Count	2300.00	2300.00	2300.00	2300.00	2300.00	2300.00
Mean	0.00	0.00	0.00	0.00	0.00	0.00
Std Dev	1.00	1.00	1.00	1.00	1.00	1.00
Min	-3.28	-2.46	-1.99	-4.69	-1.99	-0.69
25%	-0.64	-0.68	-0.69	-0.37	-0.85	-0.64
Median	0.27	-0.14	-0.21	-0.11	0.01	-0.51
75%	0.79	0.57	0.50	0.27	0.86	-0.09
Max	2.08	2.25	2.48	5.11	1.74	2.15
Kurtosis	0.59	-0.47	-0.30	5.76	-1.05	-0.04
Skewness	-0.90	0.39	0.69	1.08	-0.06	1.34

Figure 9: Kernel density for the covariates used in the damage model



5.4 Geospatial data

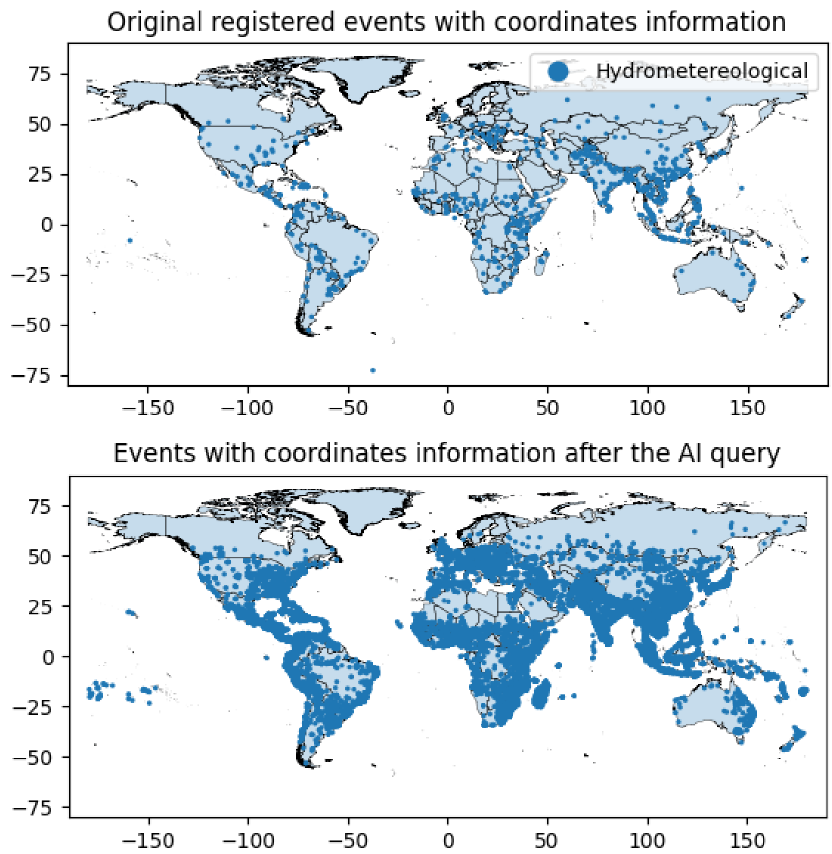
The backbone of the geospatial analysis is given by the World Bank country shapefiles (The World Bank, 2024a). Using the shapefiles, we can match geospatial information like latitude and longitude, with country-level information.

Use of Artificial Intelligence to complete locations

The use of EM-DAT disaster data at the geospatial level faces a significant challenge: out of 5,692 hydrometeorological disasters for which we have information, only 864 have associated geospatial coordinates. In contrast, 5,640 observations include data in the "Location" field, which typically contains the names of the places affected by the event in text format. It is important to note that this field often includes multiple locations, as a single disaster can impact several areas simultaneously. The key challenge, therefore, is to convert this location information from text format into geospatial coordinates.

To address this, we developed an automated query system that sends the location information to a large language model—in this case, OpenAI's GPT-4 mini (OpenAI, 2024)—and requests the corresponding geographic coordinates. We then apply a validation step to ensure that the returned coordinates fall within the correct country boundaries. After processing, we are able to transform the textual location data into 2,876 individual coordinate points associated with the recorded events. Figure 10 compares in a map the geospatial representation of disaster locations before and after the query.

Figure 10: Disasters with coordinates before and after the AI query



Figures 11 and 12 present the maps for Costa Rica and Lao with the exact location and year of the hydrometeorological disasters after the query done with Artificial Intelligence.

Figure 11: Disaster geographical distribution by year for Lao

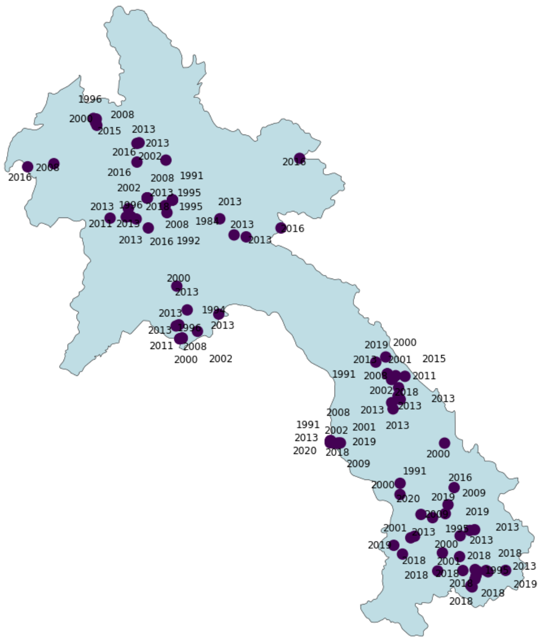
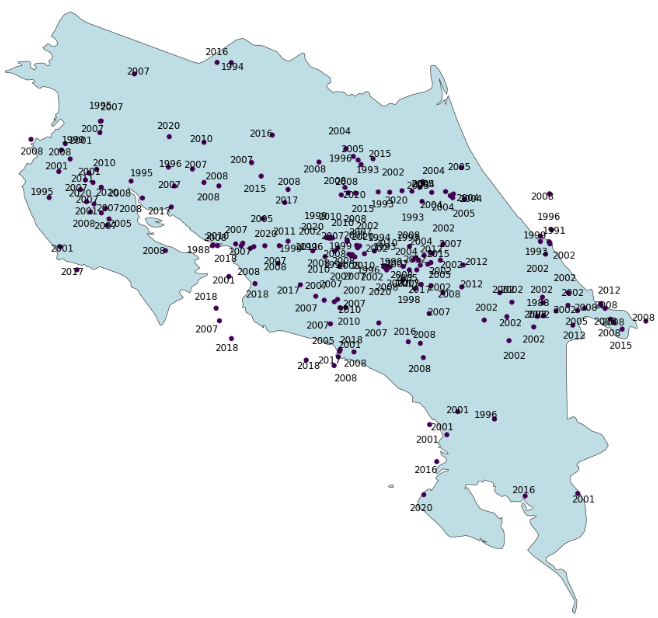


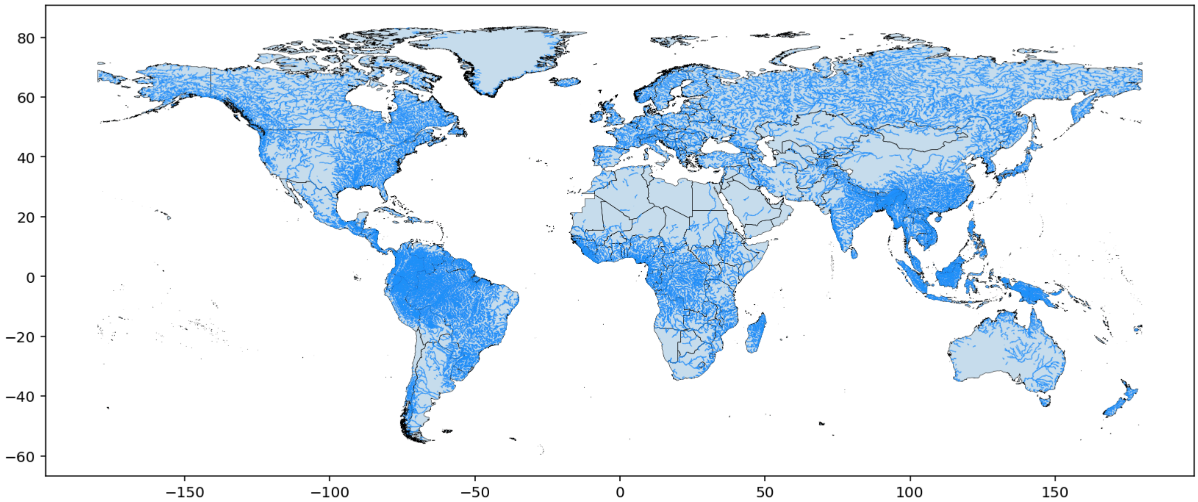
Figure 12: Disaster geographical distribution by year for Costa Rica



River data set

For the rivers, we use a similar approach. From the HydroSHEDS database (Lehner and Grill, 2013), we obtain the coordinates for all the rivers around the world. Using one criterion, we filter the rivers to have those with the greater probability of causing floods: we include only the rivers with a long-term average discharge greater than 100 m^3. Which refers to rivers that have in the data set the ORD_FLOW variable smaller than 5. Figure 13 presents the world mapping of the river data set used.

Figure 13: World rivers with a long-term average discharge greater than 100 m^3



5.4.1 Local data sets for geospatial modeling

In contrast to the first two models, which are estimated simultaneously using data from all countries with available observations, the geospatial component is estimated using data from a specific region only, and predictions are generated for a single target country. This approach is necessary due to the substantial computational resources required to estimate the model at a global scale. To illustrate the methodology, we present results for two countries: Laos and Costa Rica.

To train the model for Laos, we use data from the following countries: Cambodia, Thailand, Laos and Vietnam. While for Costa Rica we use data from Costa Rica, El Salvador, Guatemala, Honduras, Nicaragua and Panama.

Generation of fake non disaster data points

To estimate the geolocated probability of a disaster occurring at a specific point, we generate a set of synthetic non-disaster points and train a model to distinguish between real disaster locations and these synthetic counterparts. For each disaster point within a country, we randomly simulate 15 non-disaster points.

It is important to note that the disaster-to-non-disaster ratio does not affect the resulting probability values, as the outputs are normalized by dividing each by the sum of all point probabilities. However, through extensive testing, we found that a 1:15 ratio is particularly effective for generating high-quality samples for the HSGP model.

We then merge the disaster and synthetic non-disaster points into a single dataset. For each point, we calculate the distance to the nearest river and the nearest coastline. We also assign to each point the relevant country-level covariates for the year in which the event occurred and we standardize them to facilitate the estimation of the model. The full set of covariates used (all standardized by subtracting the mean and scaling by the standard deviation) is:

- Log of distance to river.
- Log of distance to coastline.
- Population.
- Accumulated atmospheric CO2 in Gigatons.
- Country's precipitation deviation from its Trend.
- Ocean temperature deviation from its Trend.
- Log of population density.
- Log of population density squared.
- Log of GDP per capita.
- Log of GDP per capita squared.

Figures 14 and 15 show the maps of disaster and synthetic non-disaster points for both regions. Since we generate 15 random non-disaster points for every disaster, the point density is lower in countries with fewer recorded disasters. It is important to note that this does not affect the model's estimates related to event frequency, as that aspect is handled by the event component of the model, not in the geospatial component described here.

In the case of Costa Rica, we exclude Isla del Coco from the map and its surroundings to facilitate visualization.

Figure 14: Disaster and synthetic non-disaster points for Central America

Yellow points correspond to reported disasters, purple to randomly generated non-disaster points.

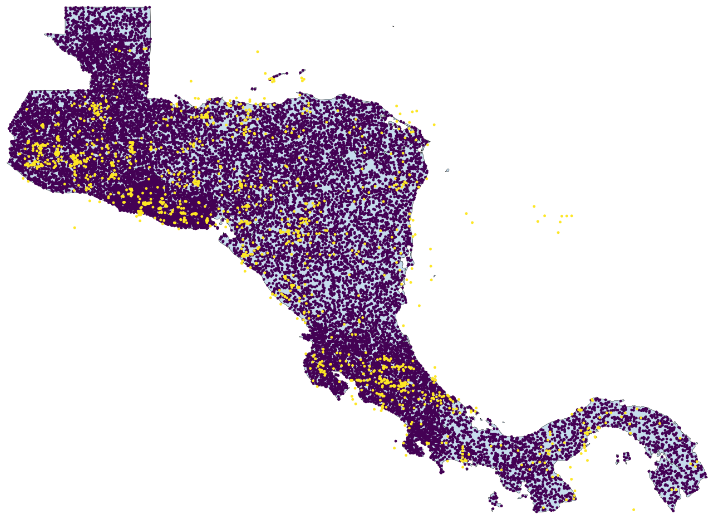
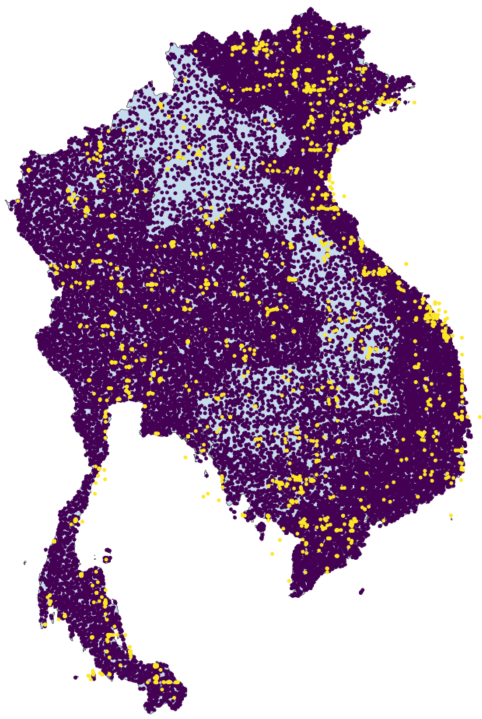


Figure 15: Disaster and synthetic non-disaster points for Cambodia, Thailand, Laos and Vietnam

Yellow points correspond to reported disasters, purple to randomly generated non-disaster points.



We provide the descriptive statistics of the datasets used to train the model for Costa Rica and Laos in Tables 6 and 7, respectively. The histograms and kernel density plots are presented in Figures 47 and 48 in Annex 1.

Table 6: Summary Statistics of the geo-spatial model data set for Central America

All features, except for "is_disaster" have been standardized.

Stat	is_dis	log_riv	log_coast	Pop	co2	prec	ocean_t	pop_dens	gdp_pc
Count	22092	22092	22092	22092	22092	22092	22092	22092	22092
Mean	0.04	-0.00	-0.00	0.00	0.00	0.00	-0.00	-0.00	0.00
Std	0.20	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Min	0.00	-10.09	-7.73	-1.31	-1.72	-1.60	-2.28	-1.59	-2.77
25%	0.00	-0.44	-0.45	-0.72	-0.77	-0.82	-0.73	-0.83	-0.68
Med	0.00	0.23	0.24	-0.30	-0.07	-0.15	-0.06	-0.05	-0.06
75%	0.00	0.68	0.74	0.53	0.56	0.76	0.94	0.80	0.64
Max	1.00	2.38	1.29	2.72	2.21	2.44	1.96	1.83	2.79
Kurt	18.78	3.95	4.13	0.36	-0.30	-0.53	-0.61	-1.13	-0.11
Skew	4.56	-1.55	-1.65	1.13	0.60	0.62	0.05	0.16	0.11

Table 7: Summary Statistics of the geo-spatial model data set for Cambodia, Thailand, Laos and Vietnam

All features, except for "is_disaster" have been standardized.

Stat	is_dis	log_riv	log_coast	Pop	co2	prec	ocean_t	pop_dens	gdp_pc
Count	30030	30030	30030	30030	30030	30030	30030	30030	30030
Mean	0.06	0.00	0.00	0.00	-0.00	-0.00	-0.00	-0.00	-0.00
Std	0.23	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Min	0.00	-7.92	-8.83	-2.35	-1.66	-2.48	-2.35	-3.44	-1.92
25%	0.00	-0.46	-0.49	-0.27	-0.88	-0.74	-0.71	-0.42	-0.81
Med	0.00	0.22	0.29	0.21	-0.07	-0.01	-0.09	-0.03	0.02
75%	0.00	0.69	0.75	0.61	0.80	0.65	0.91	0.82	0.82
Max	1.00	1.66	1.30	1.49	1.86	2.81	1.95	1.28	1.62
Kurt	13.13	3.35	3.36	0.44	-1.13	-0.00	-0.46	1.52	-1.11
Skew	3.89	-1.46	-1.57	-1.05	0.20	0.40	-0.01	-1.17	-0.17

5.5 Result grids

To visualize the model results and evaluate the probability of a disaster happening on every point of Costa Rica and Laos, we project each country's map to a 400 x 400 grid. Then for each point of the grid we compute the distance to the closest river, and the distance to the nearest coastline point, then we merge that grid data set with the country level indicators of the year to evaluate. We present the grids for both countries, and the plotting of the distance to rivers and coastline, in Figures 16 and 17.

Figure 16: Costa Rica grid with distance to rivers and coastlines

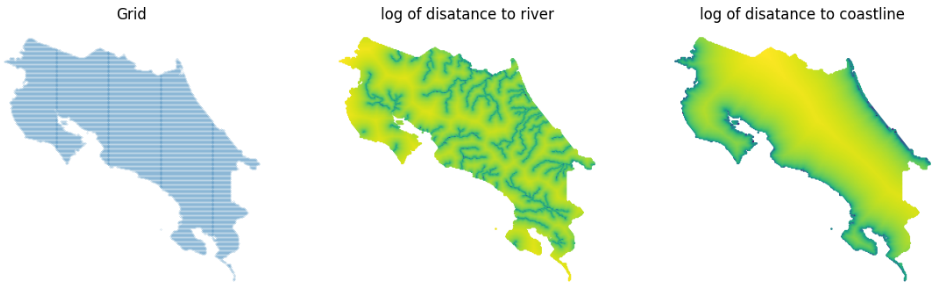
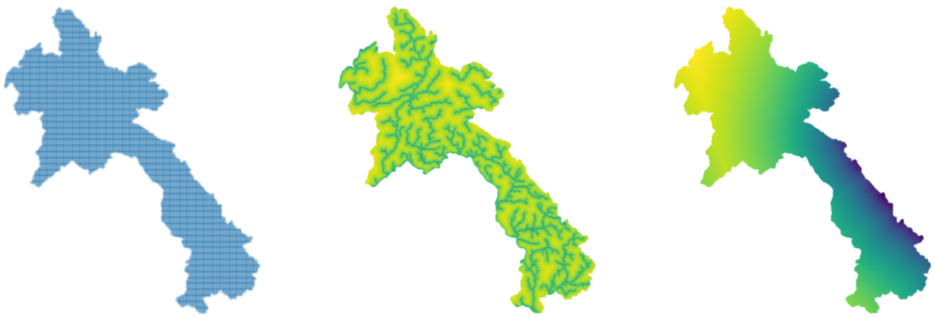


Figure 17: Laos grid with distance to rivers and coastlines



5.6 Climate and development time series

We model two types of time series: climate and development indicators in order to obtain future projections.

For the climate ones, we use:

- Total atmospheric CO_2 accumulation (in gigatons) (NOAA, 2024)
- Mean ocean temperature (Zhang et al., 2019).
- Country level mean monthly precipitation (Schneider, 2022).

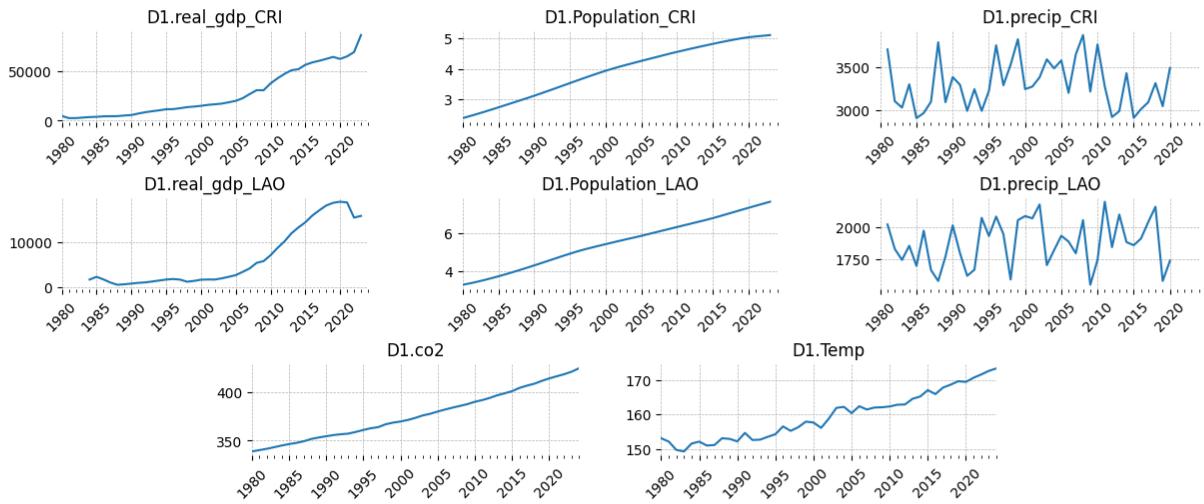
For the country development indicators, we extract from the World Development Indicators (The World Bank, 2024*b*):

- Real GDP in USD millions.
- Population in millions.

Figure 18 shows the temporal evolution of the variables included in the time series analysis. To deal with the trends and facilitate estimation, we standardize the data and compute the series first differences. Figure 50 in the annex presents the plots of this transformed data. The descriptive statistics for the transformed time series and their respective kernel density plots are presented in Table 15 and Figure 49 respectively. It is necessary to comment that the Exponential Trend Smoothing (ETS) model used does not require the time series to be stationary.

Figure 18: Time series evolution through time

Population is measured in millions, real GDP in millions of USD, CO2 in Gigatons, ocean temperature deviation from its mean in Fahrenheit degrees, and precipitation deviation is the whole year monthly mean in millimeters.



6 Results

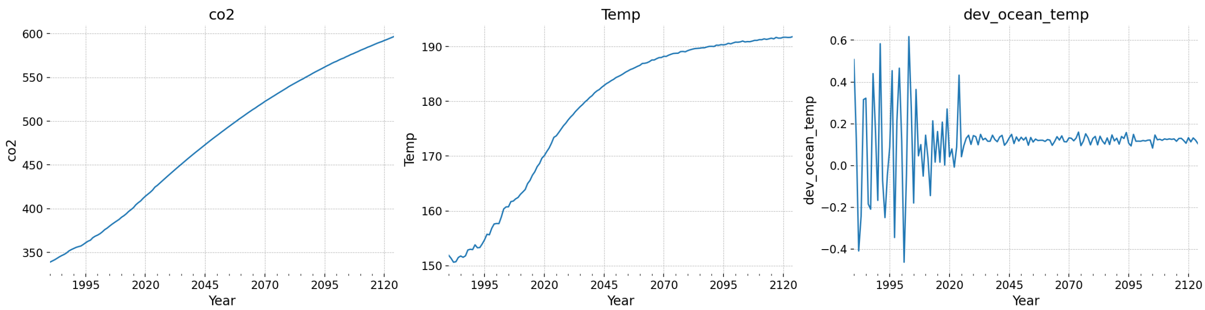
As a first step we present the results of the time series models for both countries. These results are extremely important as they allow us to project the covariates of the other models into the future, which is a requisite to produce predictions of the model results in the years to come.

6.1 Time series model results

The goal of the time series model is to generate predictions for the covariates used in the disaster frequency, disaster damages, and geospatial patterns models. This allows us to project the models into the future and evaluate the impact of different CO_2 scenarios on country-level damage curves. First, we present the results for the global-level time series of CO_2 and ocean temperature, along with the deviation of ocean temperature from its trend. We observe that both atmospheric CO_2 and ocean temperature are projected to continue their upward trends, though the growth rate of ocean temperature is expected to smooth over time.

Figure 19: Time series predictions for the global-level covariates

CO2 is measured in Gigatons, ocean temperature is measured in Fahrenheit degrees, and ocean temperature deviation from its mean is measured in standard deviation units.



In the case of CO_2 emissions, we are interested in contrasting our projections with those developed by the IPCC (Intergovernmental Panel on Climate Change). The IPCC created several Special Report on Emissions Scenarios (SRES) (Nakicenovic et al., 2000) to model possible future climate conditions based on different socioeconomic pathways. Among these, A2 and B1 represent contrasting scenarios of global development and CO_2 emissions. Scenario A2 describes a high CO_2 emissions trajectory, often considered conservative in its assumptions, while scenario B1 represents an optimistic low CO_2 emissions pathway. Table 8 shows the comparison between these two scenarios.

Table 8: Comparison of IPCC Scenarios A2 and B1

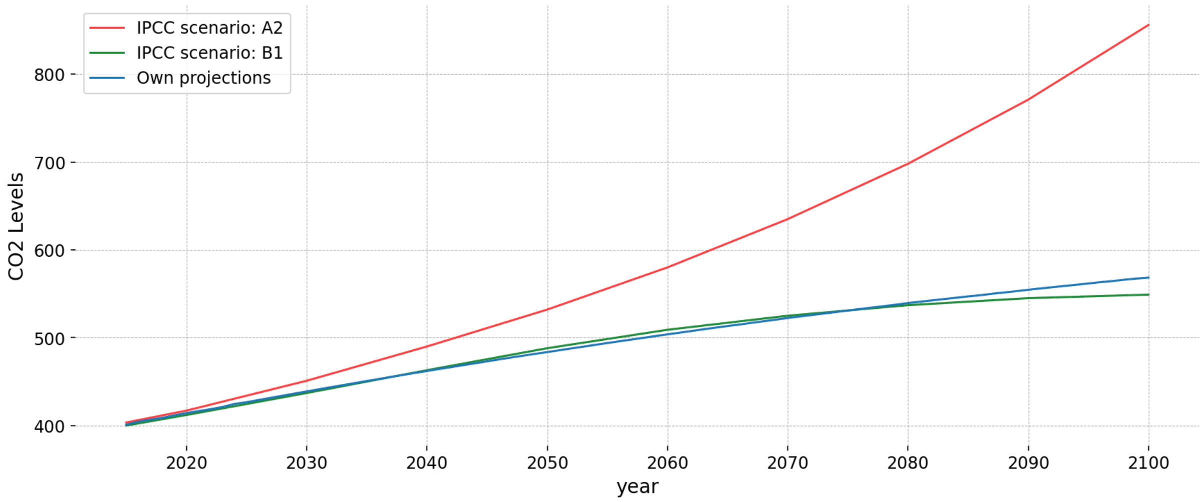
This table compares key factors between the high emissions (A2) and low emissions (B1) scenarios.

Factor	A2 (High Emissions)	B1 (Low Emissions)
Economic Growth	Regional, slower progress	Global, sustainable
Population	∼15 billion (high)	∼9 billion (stable)
Energy	Fossil fuel-dominated	Renewable energy transition
CO_2 Levels (2100)	∼850+ ppm	∼550 ppm
Temperature Rise (2100)	3.5 - 5.5°C	1.5 - 2.5°C
Climate Risks	Severe heatwaves, storms, sea level rise	Milder impacts, adaptation possible

Figure 20 compares the trajectory of IPCC scenarios A2 and B1 with our own projections. We observe that scenario B1 aligns relatively closely with what our time series model projects, while scenario A2 shows a significant divergence. Our focus is on understanding how different CO_2 trajectories might impact the damage curves for the countries analyzed. To address this question, we will model the damage curves using both scenario A2 and our own projections.

Figure 20: IPCC and own projections of atmospheric CO_2

CO2 is measured in Gigatons, while ocean temperature is measured in Fahrenheit degrees. The deviation of ocean temperature from its trend value is measured in standard deviations.



Regarding the country-level variables, Figures 21 and 22 present the projections for Costa Rica and Laos, respectively. In both cases, we observe an upward trend in GDP and population. For Costa Rica, the GDP growth rate surpasses the population growth rate, resulting in a rising GDP per capita. In contrast, for Laos, the projected population growth rate exceeds the GDP growth rate during a certain period, leading to a decline in GDP per capita. In the case of precipitation, the time series for both countries stabilize after following the current trend for a few years.

Figure 21: Time series predictions for Costa Rica country level variables

Population is measured in millions, real GDP in millions of USD, and precipitation is the whole monthly mean in millimeters.

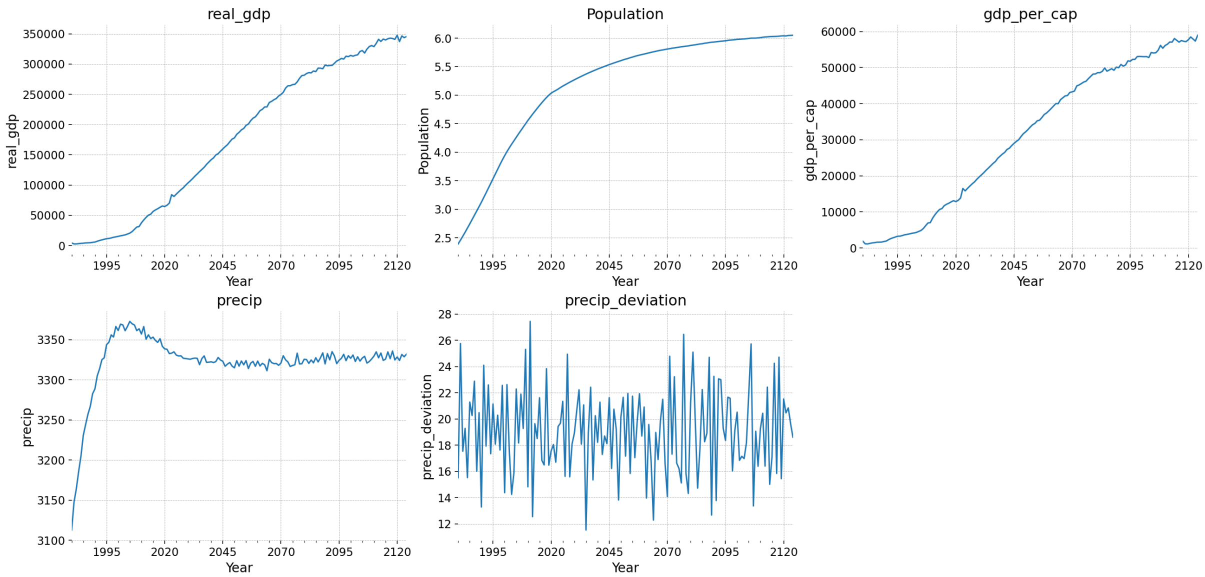
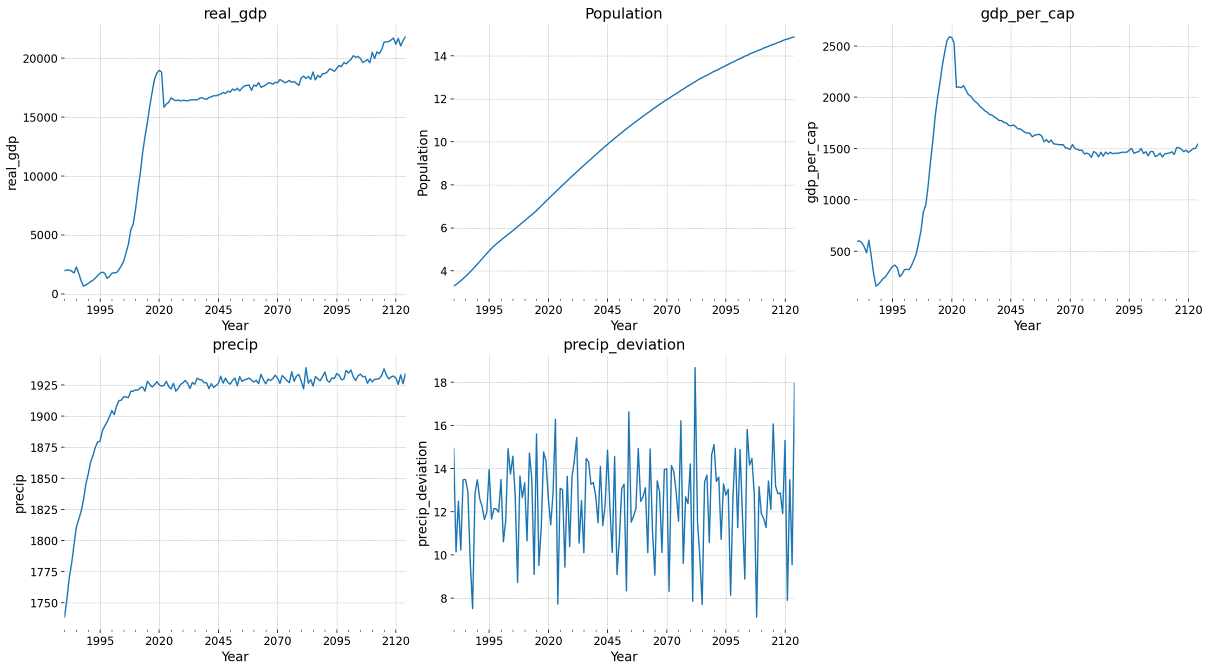


Figure 22: Time series predictions for Laos country level variables

Population is measured in millions, real GDP in millions of USD, and precipitation is the whole monthly mean in millimeters.



6.2 Disaster frequency model

The sampling statistics of the frequency model are presented on tables 8 and 9. The sampling was implemented without divergences. The r_hat values for the results show a healthy sampling process. According to the Gelman–Rubin convergence diagnostic, the values of 1.00 mean that MCMC chains have converged to the same target posterior distribution. The only exception is the case of beta_GY, which has some r_hat values above 1.00. In general, the ess_tail, which measures the effective sample size in the tails of the posterior distribution, reveal that we have reasonable sampling sizes, with the lowest one being 945.

Table 9: Summary Statistics 1 for frequency event model estimation

Variables contained: alpha and alpha_X.

Parameter	mean	sd	hdi_3%	hdi_97%	ess_tail	r_hat
alpha	10.923	1.575	8.076	13.839	6169.0	1.0
alpha_X[ln_pop_density]	0.002	2.400	-4.666	4.614	6163.0	1.0
alpha_X[ln_gdp_pc]	-0.047	2.429	-4.824	4.463	6167.0	1.0
alpha_X[precip_deviation]	0.002	2.351	-4.368	4.521	6562.0	1.0
alpha_X[co2]	-0.006	2.405	-4.545	4.826	6275.0	1.0
alpha_X[population]	0.043	2.522	-4.605	4.951	6086.0	1.0
alpha_X_loc[ln_pop_density]	-0.019	1.007	-1.927	1.847	5732.0	1.0
alpha_X_loc[ln_gdp_pc]	-0.003	1.005	-1.897	1.903	5927.0	1.0
alpha_X_loc[precip_deviation]	0.000	0.981	-1.833	1.844	6378.0	1.0
alpha_X_loc[co2]	-0.001	1.008	-1.909	1.881	5888.0	1.0
alpha_X_loc[population]	0.014	1.000	-1.777	1.943	6242.0	1.0
alpha_X_offset[ln_pop_density]	0.008	0.907	-1.770	1.643	5903.0	1.0
alpha_X_offset[ln_gdp_pc]	-0.004	0.898	-1.637	1.729	6051.0	1.0
alpha_X_offset[precip_deviation]	-0.007	0.886	-1.671	1.692	6037.0	1.0
alpha_X_offset[co2]	-0.006	0.885	-1.650	1.661	5974.0	1.0
alpha_X_offset[population]	0.010	0.929	-1.729	1.739	5775.0	1.0
alpha_X_scale[ln_pop_density]	1.996	1.416	0.037	4.517	4929.0	1.0
alpha_X_scale[ln_gdp_pc]	2.001	1.433	0.051	4.544	5176.0	1.0
alpha_X_scale[precip_deviation]	1.979	1.398	0.034	4.452	5291.0	1.0
alpha_X_scale[co2]	1.989	1.410	0.051	4.476	4705.0	1.0
alpha_X_scale[population]	2.002	1.418	0.033	4.465	4370.0	1.0

Figure 23 presents the model results for Lao. Here the blue line provides the mean in-sample prediction, while the shaded region corresponds to the 95% High-Density-Interval (the darker the shade, the darker the probability for that value). We can see how the model succeeds to capture the general trend of the event's frequency, and how almost the observed number of events lie inside the 95% HDI values. In general, Lao presents a mean yearly number of events lower than one, with a trend to grow during the last decade, where the mean number of events increases up to one.

Figure 24 reveals a clear upward trend in the number of yearly events, with the mean value increasing up to 3. The 95% Highest Density Interval (HDI) extends up to 6 events. This means that by 2020, we could be 95% confident that the number of hydrometeorological disasters in Laos was fewer than 3. By 2060, however, this number could increase to 6. (It is important to highlight that, for the purposes of this study, we are only considering events reported to affect more than 1,000 people).

Table 10: **Summary Statistics 2 for frequency event model estimation**

Statistic	mean	sd	hdi_3%	hdi_97%	ess_tail	r_hat
beta[ln_pop_density]	0.338	0.108	0.140	0.544	4237.0	1.00
beta[ln_gdp_pc]	-0.651	0.088	-0.813	-0.478	3745.0	1.00
beta[precip_deviation]	0.002	0.000	0.002	0.003	6027.0	1.00
beta[co2]	0.025	0.002	0.022	0.029	4213.0	1.00
beta[population]	0.010	0.001	0.008	0.012	5081.0	1.00
beta_GY[ln_pop_density]	-0.548	0.306	-1.119	0.010	1263.0	1.01
beta_GY[ln_gdp_pc]	-0.559	0.298	-1.120	-0.001	945.0	1.01
beta_GY[precip_deviation]	-0.574	0.307	-1.143	-0.010	1017.0	1.03
beta_GY[co2]	-0.526	0.305	-1.092	0.048	846.0	1.02
beta_GY[population]	-0.550	0.295	-1.074	0.027	1468.0	1.01
country_loc	-3.935	0.644	-5.203	-2.784	4829.0	1.00
country_scale	0.340	0.193	0.024	0.686	3321.0	1.00
rho	0.878	0.138	0.595	1.000	1132.0	1.01
sigma_X[ln_pop_density]	1.004	0.993	0.000	2.865	3529.0	1.00
sigma_X[ln_gdp_pc]	1.002	1.016	0.000	2.843	3382.0	1.00
sigma_X[precip_deviation]	1.006	1.010	0.000	2.868	3393.0	1.00
sigma_X[co2]	0.996	0.984	0.000	2.777	3547.0	1.00
sigma_X[population]	1.006	1.023	0.000	2.896	3314.0	1.00
zero_prob	0.998	0.002	0.994	1.000	3568.0	1.00

Figure 23: Event Frequency model results for Lao

Black points correspond to the observed numbers, blue line to the in-sample mean prediction, and shaded region to the 95% HDI, where lighter shadow reflects lower probability.

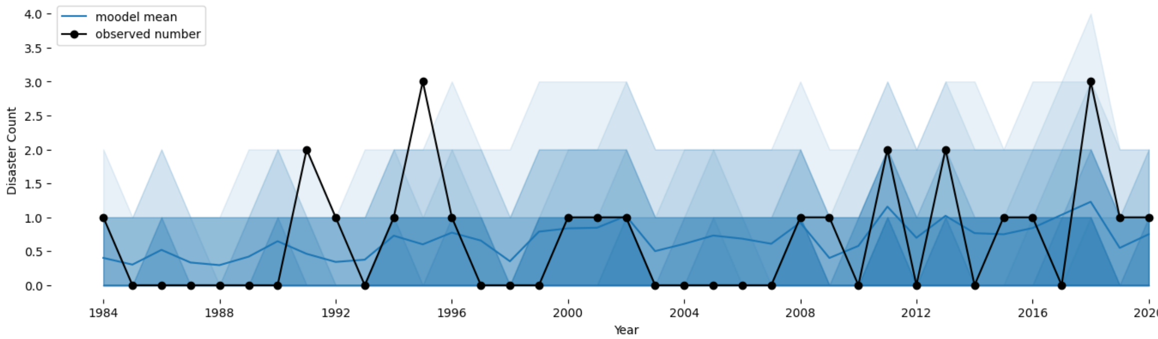


Figure 24: Event Frequency predictions for Lao

Blue line to the mean prediction, and shaded region to the 95% HDI, where lighter shadow reflects lower probability.

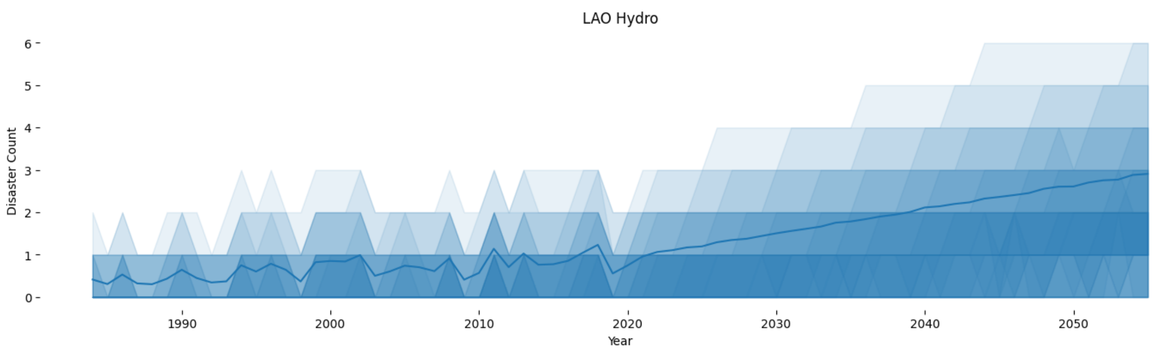


Figure 25 presents the model results for Costa Rica. The mean number of yearly disasters in the country starts around 0.5, with a growing trend that makes it approach one. It is worth noting that in 2008 Costa Rica registered 4 events in one single year. In general we observe that the 95% High Density Interval manages to capture all the yearly values.

Figure 25: Event Frequency model results for Costa Rica

Black points correspond to the observed numbers, blue line to the in-sample mean prediction, and shaded region to the 95% HDI.

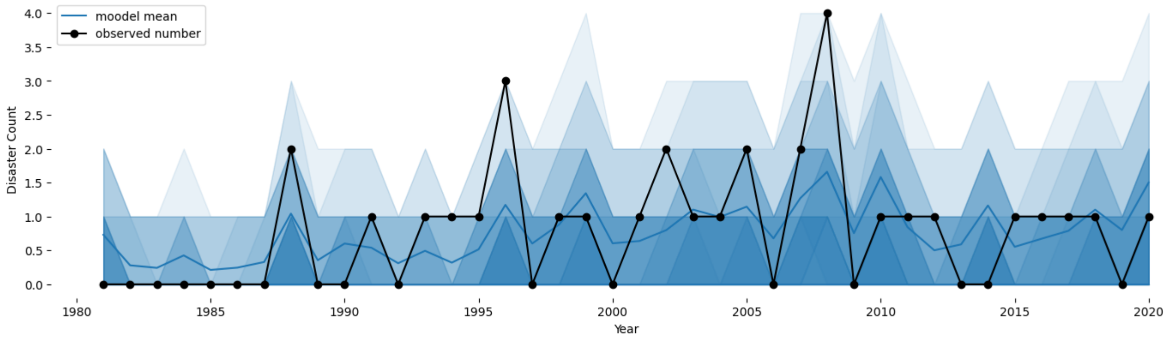
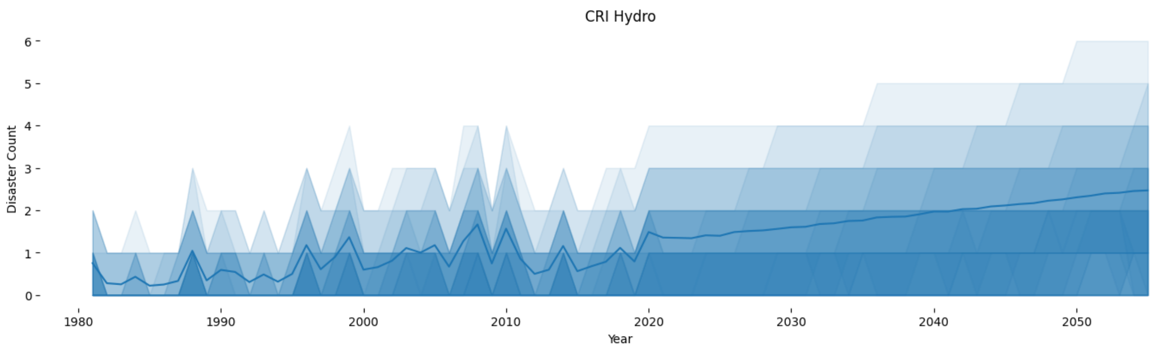


Figure 26 presents the projected trends. Just like in the case of Laos, we observe a growing trend, nevertheless in the case of Costa Rica, with the mean value reaching 2.5 in 2060, and the upper band of the 95% HDI growing up to 6 yearly disasters.

Figure 26: Event Frequency predictions for Costa Rica

Black points correspond to the observed numbers, blue line to the in-sample mean prediction, and shaded region to the 95% HDI.



We can conclude then that both countries are exposed to an increase of the number of hydrometeorological events in the years to come according to our model.

6.3 Damages model

The sampling results for the damage model are presented on table 10, while the in sample model predictions are presented on Figures 27 and 28. The model was successfully sampled without divergences. From table 10 it is possible to observe that almost all variables have a r_hat of 1.00 which reveals a healthy posterior approximation. However three particular variables have r_hat values greater to 1.01: alpha_X[ln_population_density], alpha_X[ln_gdp_pc], and beta_GX[ln_population_density]. This reveals the MCMC had some conversion troubles for those particular variables. However, all the variables remain under the 1.05 threshold.

The effective sample sizes (ess_tail) are generally high, indicating that the MCMC chains have mixed well and that the posterior distributions are well-characterized. The variables alpha_X[ln_population_density], alpha_X[ln_gdp_pc], and beta_GX[ln_population_density] present the lowest ess_tails, however the values are all equal or greater to 280, which still represent a reasonable size for posterior drawing.

Table 11: Summary Statistics of Model Parameters

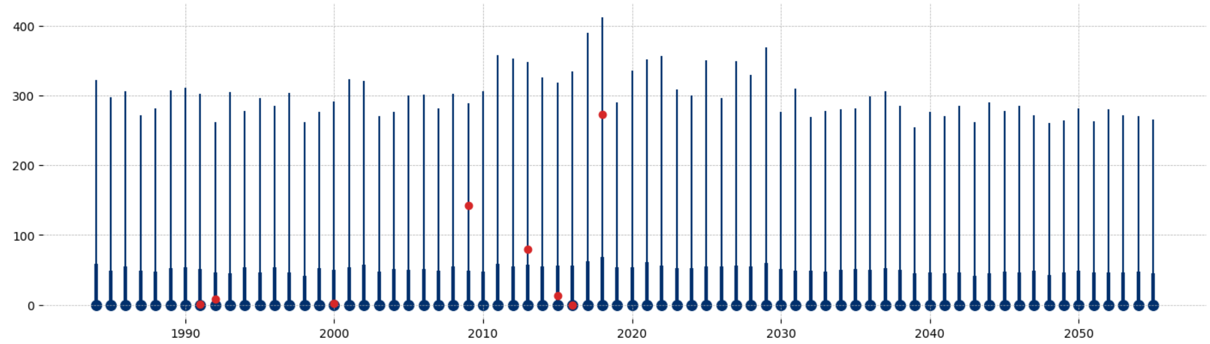
Parameter	mean	sd	hdi_3%	hdi_97%	ess_tail	r_hat
alpha_X[ln_population_density]	-0.470	0.077	-0.623	-0.332	280.0	1.02
alpha_X[ln_gdp_pc]	-0.087	0.073	-0.224	0.052	376.0	1.02
alpha_X[precip_deviation]	0.621	0.086	0.464	0.781	1489.0	1.00
alpha_X[co2]	0.524	0.086	0.363	0.680	1366.0	1.00
alpha_X[population]	-0.590	0.075	-0.730	-0.447	544.0	1.01
beta[ln_population_density]	-1.047	0.494	-1.898	-0.057	3479.0	1.00
beta[ln_gdp_pc]	0.413	0.185	0.062	0.760	3540.0	1.00
beta[precip_deviation]	0.076	0.036	0.008	0.142	2549.0	1.00
beta[co2]	0.040	0.079	-0.111	0.184	3116.0	1.00
beta[population]	-0.075	0.300	-0.619	0.511	3245.0	1.00
beta_GX[ln_population_density]	0.308	0.010	0.288	0.328	688.0	1.03
beta_GX[ln_gdp_pc]	-0.007	3.109	-5.797	5.695	2366.0	1.00
beta_GX[precip_deviation]	-0.007	3.052	-5.386	6.063	2445.0	1.00
beta_GX[co2]	0.010	3.015	-5.790	5.442	2686.0	1.00
beta_GX[population]	0.028	2.993	-5.584	5.513	3067.0	1.00
beta_GY[ln_population_density]	0.282	0.256	-0.210	0.753	3415.0	1.00
beta_GY[ln_gdp_pc]	-0.056	0.231	-0.503	0.375	3005.0	1.00
beta_GY[precip_deviation]	-0.169	0.230	-0.635	0.235	2889.0	1.00
beta_GY[co2]	-0.106	0.235	-0.552	0.332	2778.0	1.00
beta_GY[population]	0.047	0.242	-0.397	0.520	2769.0	1.00
country_loc	2.616	0.301	2.062	3.185	2102.0	1.00
country_scale	0.145	0.118	0.000	0.367	1814.0	1.01
p_zero	0.002	0.001	0.000	0.004	2660.0	1.00
rho	0.959	0.055	0.861	1.000	1224.0	1.00
sigma	2.164	0.033	2.098	2.223	3026.0	1.00
sigma_X[ln_population_density]	0.125	0.002	0.121	0.128	2441.0	1.00
sigma_X[ln_gdp_pc]	0.338	0.005	0.327	0.347	2674.0	1.00
sigma_X[precip_deviation]	1.258	0.029	1.205	1.312	3285.0	1.00
sigma_X[co2]	1.259	0.029	1.208	1.314	3362.0	1.00
sigma_X[population]	0.126	0.002	0.123	0.130	2733.0	1.00

The damage model results and predictions for Lao are presented in Figure 27. This plot shows the in-sample damage predictions conditional on the occurrence of a disaster. We can observe how the country has experienced in the last decade three disasters that surpass in terms of damage all previous disasters. In that regard the 2018 flood was particularly destructive, with damages outside the 80% HDI. Interestingly, all the reported events are contained in the 95% HDI, with only two events outside the 80% HDI.

In terms of the projected values, the model does not project a growing trend in terms of the projected damages. Here it is important to take into consideration that the model has country-level hierarchy for different parameters, which means it infers different relations between covariates for countries.

Figure 27: Damage model results for Laos

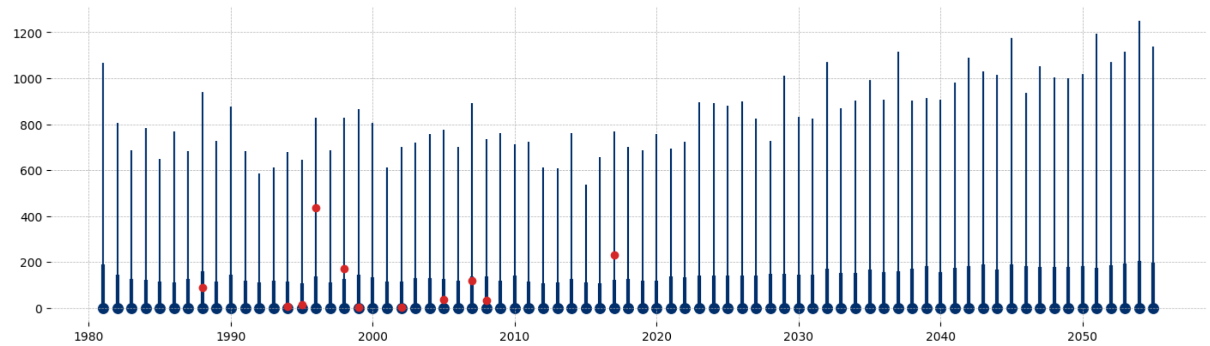
Red points correspond to the observed damages, light blue line corresponds to the 97.5% HDI, and wider blue line corresponds to 90% HDI. Damages are expressed in 2025 USD millions.



Results for Costa Rica are presented in Figure 28. All the registered events lie inside the 97.5% HDI. The two more destructive events correspond to the impact of Hurricane Cesar in 1996 and Hurricane Otto in 2017. Those are the only events outside the 80% HDI. In the case of the Central American country we do observe an increasing trend for the projected damages. With the upper band of the 97.5% of the High Density Interval increasing from 800 USD millions in 2020 to almost 1100 USD millions by 2060.

Figure 28: Damage model results for Costa Rica

Red points correspond to the observed damages, light blue line corresponds to the 95% HDI, and wider blue line corresponds to 90% HDI. Damages are expressed in 2025 USD millions.

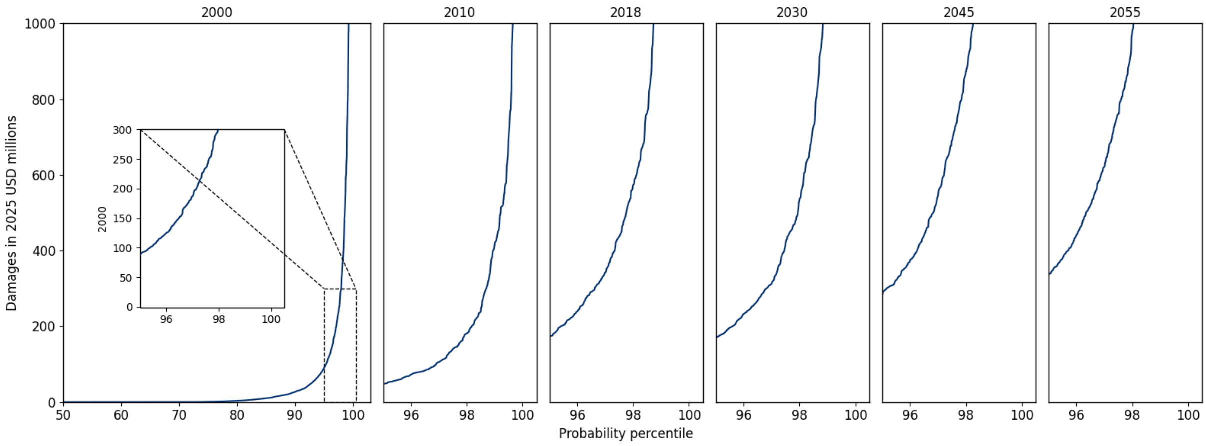


Therefore, our model predicts that Costa Rica will have a greater exposure to more impactful events in the coming decades, while the projected mean damage of each event is not projected to increase in the case of Laos. Nevertheless, it is necessary to underscore that the total exposure is measured by the interaction of damages and frequency, which we will explore in the next section.

6.4 Country level damages curve

To produce the damage curves, we combine the results of the damage and frequency models. Figure 29 presents the damage curves for Laos. The vertical axis corresponds to damages in millions of USD, while the horizontal axis represents the probability associated with each damage level. As expected, most of the probability mass corresponds to low-damage events. However, as we consider less probable disasters, the magnitude of damages increases. The most relevant finding is that the model projects an upward shift in Laos' damage curve, indicating that more impactful events will become more frequent, and more frequent events are projected to generate greater damages.

Figure 29: Lao damage curves



A more intuitive way of presenting the damage curves corresponds to the idea of **return years**. Table 12 and Figure 30 present the projected year returns for Lao. The projected return year values illustrate a clear escalation of climate-related disaster risks for Laos over time. While the frequency of smaller-scale events (e.g., 2-year and 1-year returns) remains negligible, the table shows a marked increase in damages for rarer, high-impact events. For instance, the 100-year return value rises from approximately USD 611 million in 2000 to over USD 2.1 billion by 2055. Similarly, the 50-year return value more than triples within the same period. This pattern underscores how the severity of extreme events is expected to intensify, placing growing economic and social pressure on the country.

Table 12: Projected Return Year Values by Year for Lao

This table presents return year values for the specified years in 2025 USD millions.

Return Years	2000	2010	2018	2030	2045	2055
100 RY	611.19	511.82	1195.83	1187.81	1623.36	2133.86
50 RY	307.86	188.15	581.92	493.46	864.37	997.77
10 RY	23.38	11.46	52.44	57.22	90.29	117.28
4 RY	0.72	0.00	3.47	5.39	10.58	13.53
2 RY	0.00	0.00	0.00	0.00	0.00	0.00
1 RY	0.00	0.00	0.00	0.00	0.00	0.00

Importantly, the results also suggest that moderate return periods (10-year and 4-year events) will increasingly contribute to the overall risk landscape. By 2055, a 10-year return event is projected to cause damages exceeding USD 117 million, compared to just USD 23 million in 2000. This shift means that not only will catastrophic events become more damaging, but recurrent disasters will also impose higher recurring costs.

Figure 30: Lao damage barplots

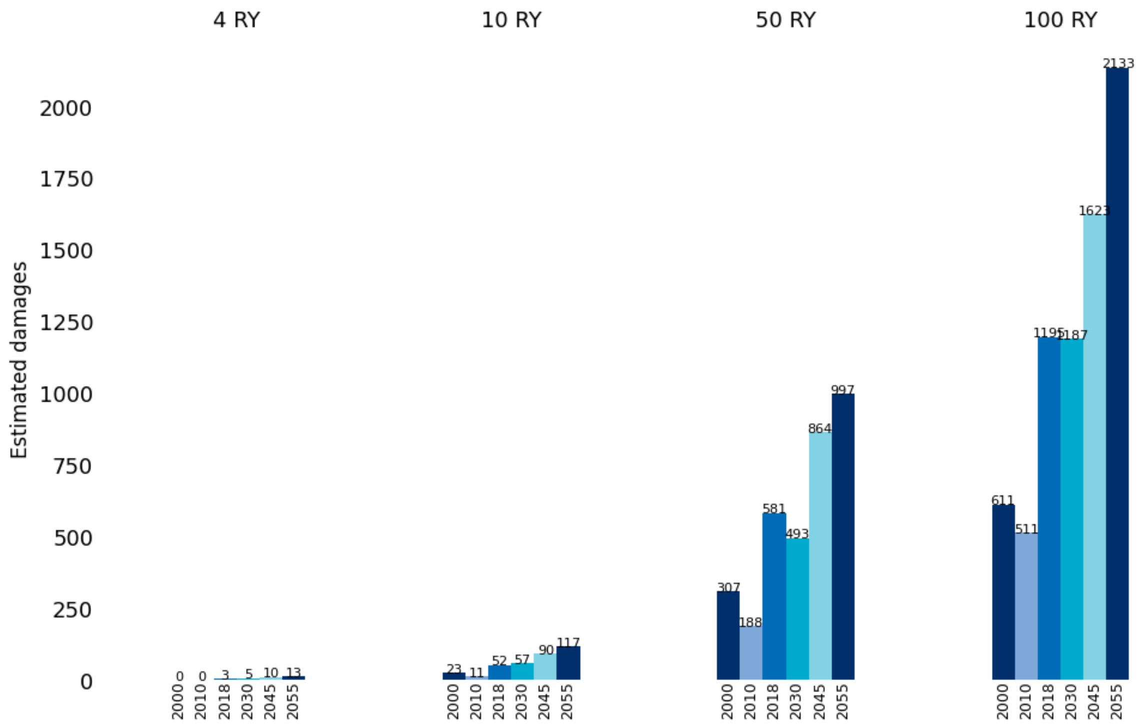


Figure 31 presents the results for Costa Rica. In this case, we observe an even more pronounced increase in projected exposure to climate change–related disasters. In 2000, the probability of a disaster causing damages of USD 1,000 million was below 1%; however, by 2055, that probability is projected to rise to nearly 5%.

Figure 31: CR damage curves

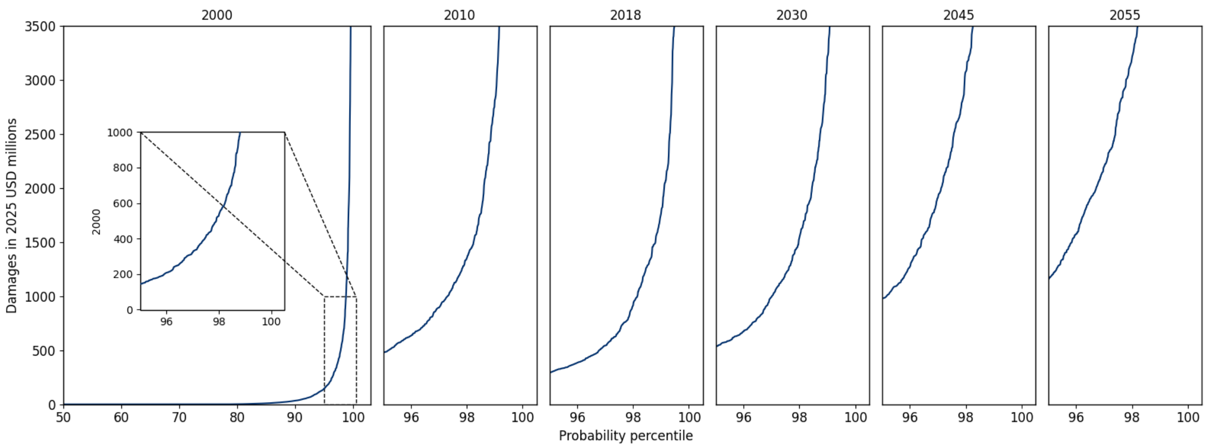


Table 13 and Figure 32 present the return-year translations of the damage curves for Costa Rica. The projected return year values reveal a steep rise in expected damages from hydrometeorological disasters over the coming decades for the Central American nation. For rare but extreme events, the increase is particularly dramatic: the 100-year return value grows from about USD 1.2 billion in 2000 to more than USD 6.5 billion by 2055. Likewise, the 50-year return damages more than quadruple over the same period. This upward trend highlights how climate change is projected to significantly amplify the economic impacts of extreme disasters.

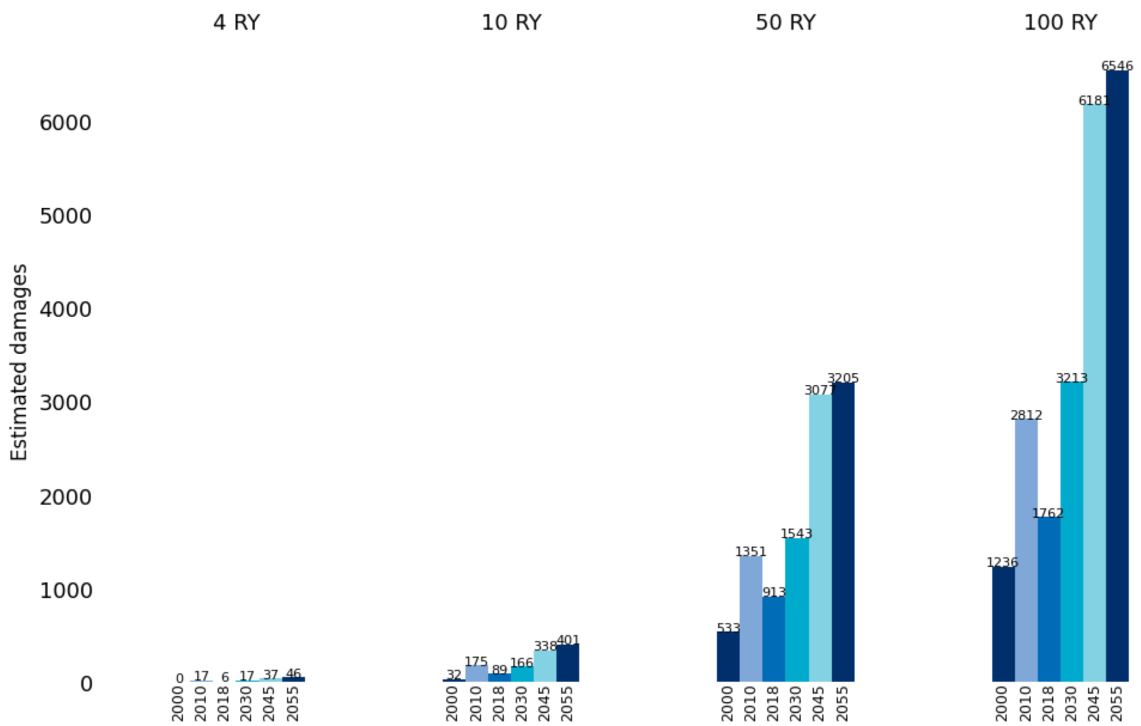
Table 13: Return Year Values by Year for Costa Rica

This table presents return year values for the specified years in 2025 USD.

Return Years	2000	2010	2018	2030	2045	2055
100 RY	1236.37	2812.63	1762.17	3213.56	6181.40	6546.61
50 RY	533.97	1351.89	913.47	1543.53	3077.38	3205.45
10 RY	32.49	175.55	89.65	166.32	338.13	401.02
4 RY	0.00	17.47	6.14	17.36	37.31	46.95
2 RY	0.00	0.00	0.00	0.00	0.00	0.00
1 RY	0.00	0.00	0.00	0.00	0.00	0.00

The data also show that damages from more frequent events are rising sharply, underscoring the increasing cost of recurrent disasters. For example, a 10-year return event is projected to escalate from about USD 32 million in 2000 to over USD 400 million by 2055, while 4-year return events—once negligible—are expected to reach nearly USD 47 million by mid-century. This shift suggests that climate change will not only increase the burden of catastrophic losses but also impose growing, recurring costs on communities and infrastructure.

Figure 32: CR damage barplots



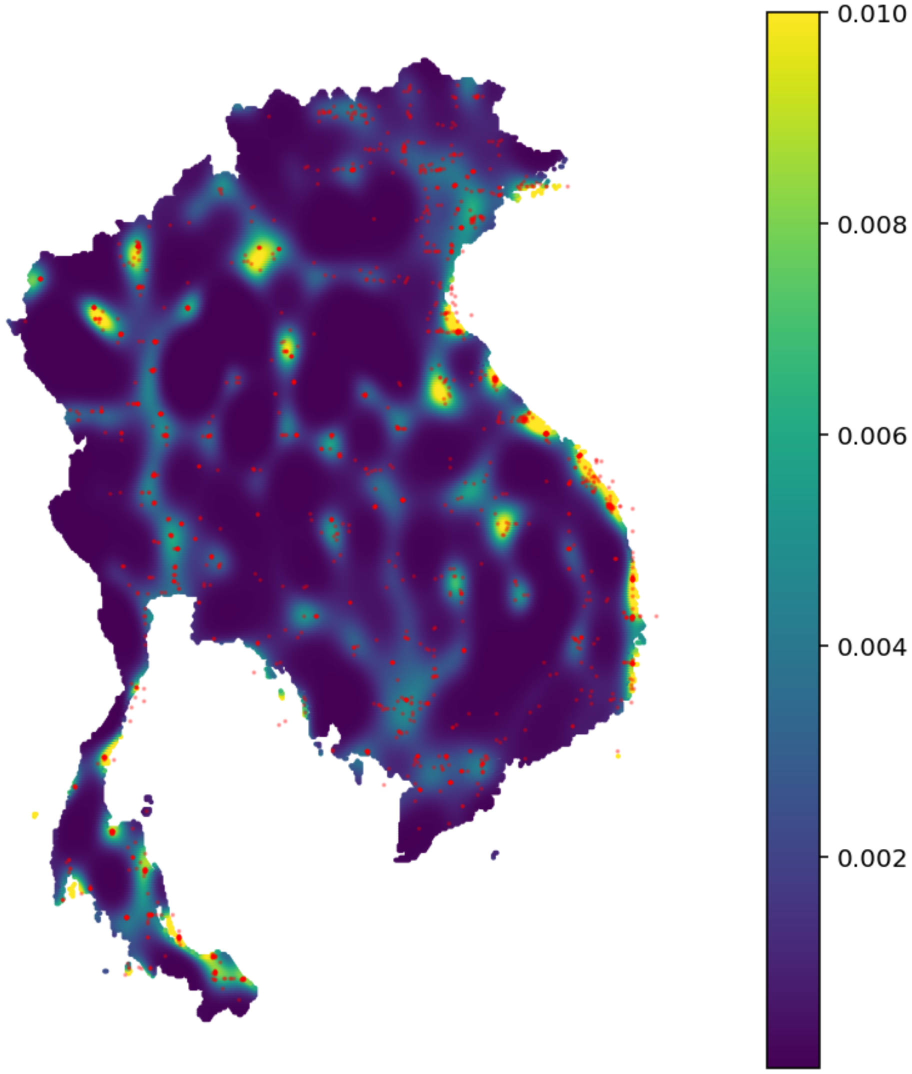
6.5 Geospatial patterns model

6.5.1 HSGP modeling

First, we implement a simplified geospatial model that relies solely on an HSGP component to estimate the probability of a disaster occurring at any given point within a country. This simplified HSGP version uses only latitude and longitude as independent variables, making it time-independent.

Figure 33: HSGP event probability predictions for Cambodia, Thailand, Laos and Vietnam

The plot value measures the yearly probability that a disaster will happen on each specific point conditional on the occurrence of one disaster.



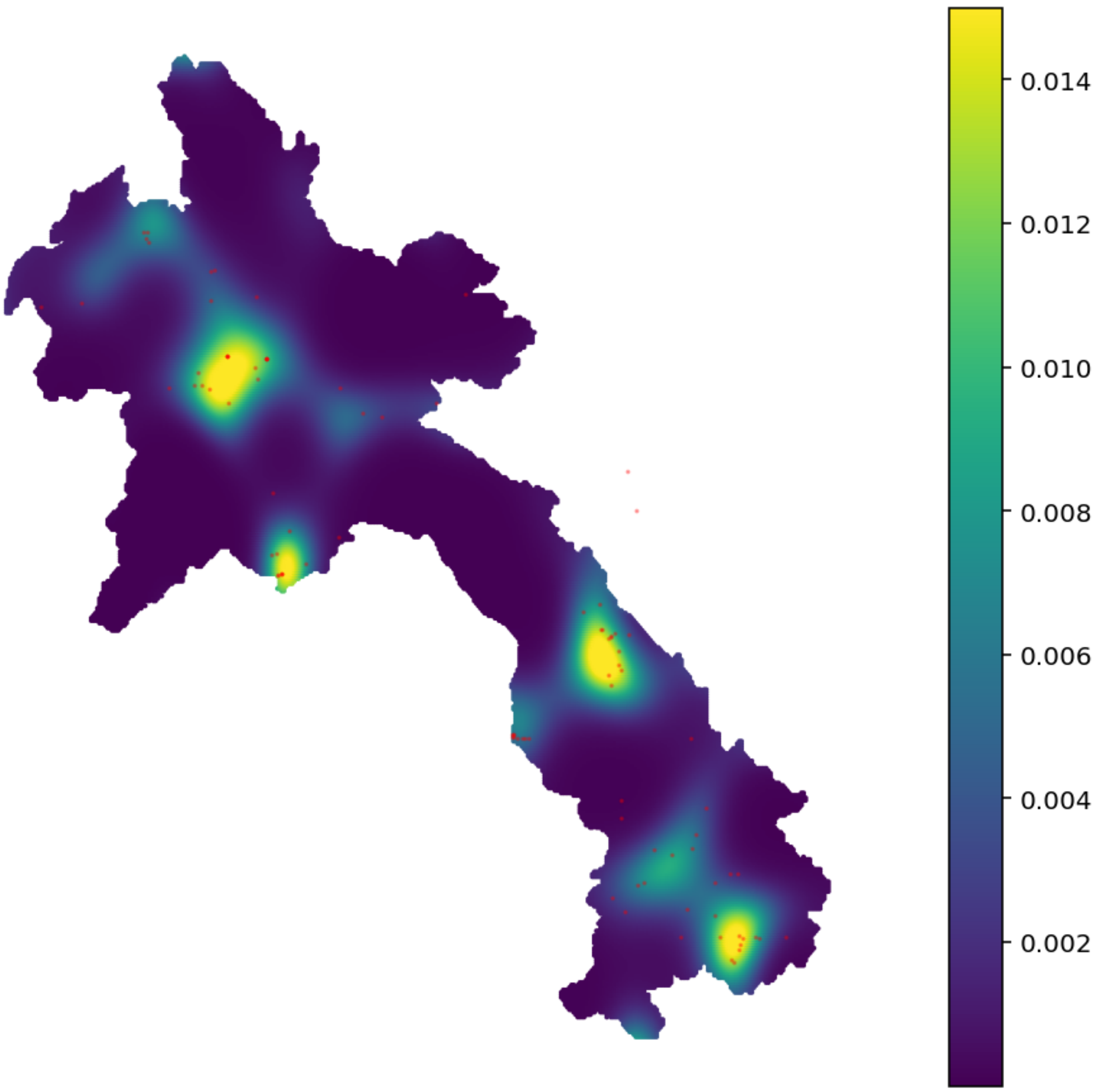
The results, shown in Figure 33, cover Cambodia, Thailand, Laos, and Vietnam. It is striking to see how the model can infer the outlines of major rivers purely from disaster occurrence patterns. We also observe that certain coastal regions exhibit greater exposure to disasters. This reinforces the idea that the HSGP component of the model represents a strong tool to account for omitted variables, as it is able to capture patterns of geographical patterns not explicitly included in the model. These results also confirm the notion that distance to rivers and coastlines represent a clear risk factor in the region.

HSGP parameter values for Cambodia, Thailand, Laos and Vietnam, and HSGP parameter values for Central America can be found on Tables 16 and 17 of Annex 2 respectively.

Figure 34 presents the results specifically for Lao. Based only on the coordinates of previous disasters, the model is able to identify a risk zone in the center-left area of the country, precisely where a big number of rivers merge. It also identifies a risk area closer to the capital Vientiane, and in two more points in the lower right area of the country.

Figure 34: HSGP event probability predictions for Lao

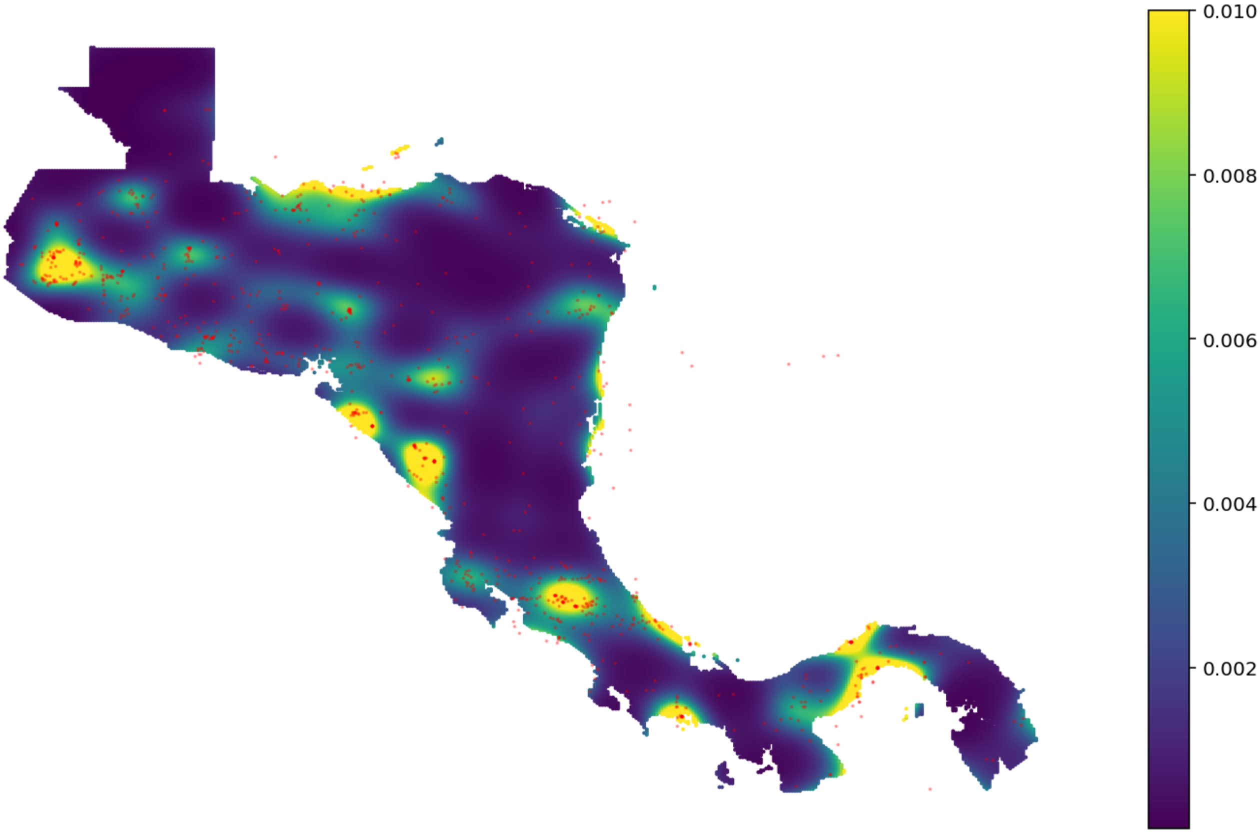
The plot value measures the yearly probability that a disaster will happen on each specific point conditional on the occurrence of one disaster.



The results for Central America are presented on Figure 35. It is possible to observe how the model successfully captures the disaster patterns.

Figure 35: HSGP event probability predictions for Central America

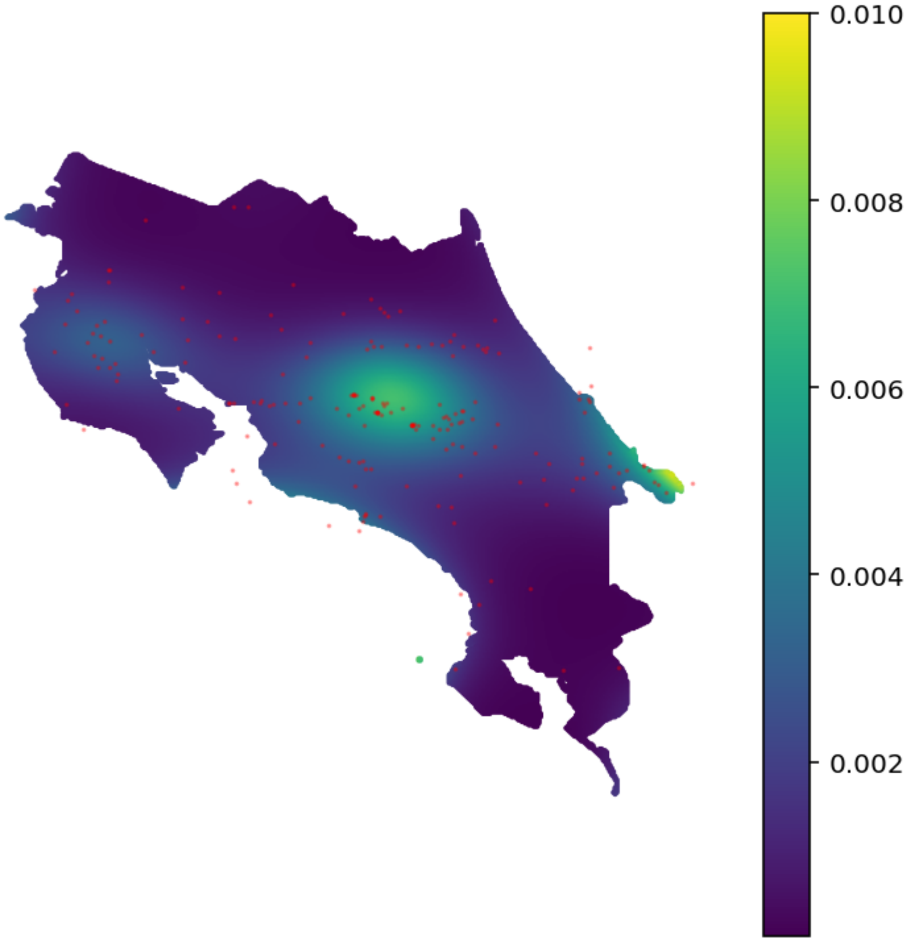
The plot value measures the yearly probability that a disaster will happen on each specific point conditional on the occurrence of one disaster.



The results for Costa Rica, presented in Figure 36, show a different pattern from those observed in Laos. The HSGP locates most of the risk in the central region of the country, known as the Greater Metropolitan Area, which is home to the majority of the population. It also indicates elevated risk in Guanacaste, particularly around Liberia, another major population center. Finally, the HSGP component estimates considerable risk along the Caribbean coast near the border with Panama, a region well known for its susceptibility to floods.

Figure 36: HSGP event probability predictions for Costa Rica

The plot value measures the yearly probability that a disaster will happen on each specific point conditional on the occurrence of one disaster.



6.5.2 Full model

Now, we present the estimated results for the full model, starting with Laos and its neighboring countries. It is important to highlight that we choose Lao, Costa Rica and their neighboring countries as examples, nevertheless, the model can generate estimations and results for almost any country or region in the world for which the data is available.

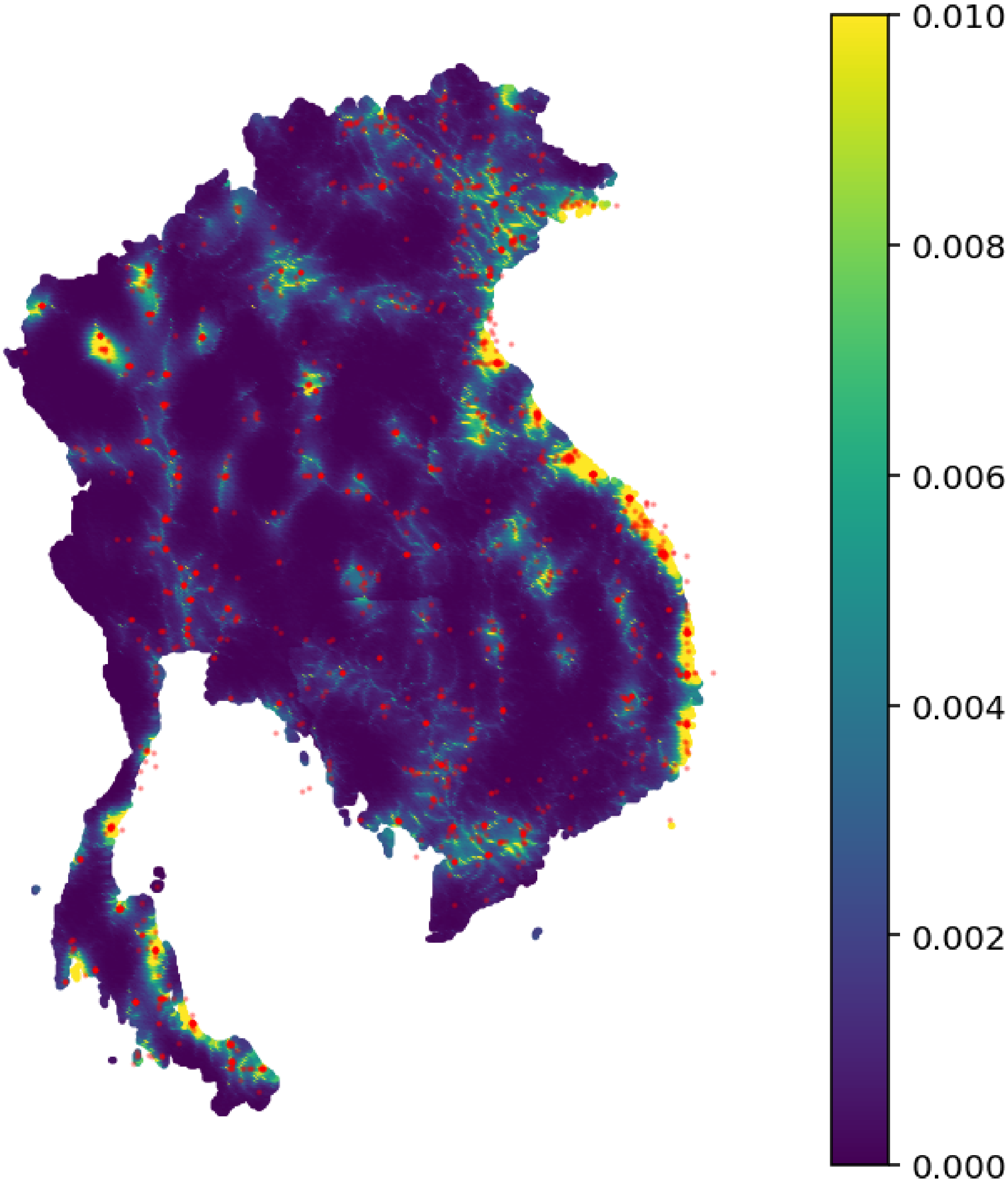
For an easier understanding, we present the results in maps, however, the model produces specific probabilities for each point of the country, which can be used for specific computations. We will present an example of this later.

Tables 18 and 19 on Annex 2 contain the main parameter values of both Full Model estimations. Figure 37 shows the results for Cambodia, Thailand, Laos, and Vietnam. It is possible to observe that the distance to rivers is a relevant determinant of exposure to hydrometeorological events. However, it is not just the distance to any river that matters—the model combines the geospatial patterns identified by the HSGP with river distance information to determine high-risk areas. Consequently, some areas have a relatively low probability of events despite being close to rivers.

The same is true for distance to the coastline. The model is able to determine which coastal regions pose the greatest risk, as can be seen in the eastern part of the region. As mentioned earlier, this is due to the capacity of the HSGP component to identify and project geospatial patterns, which helps account for missing covariates.

Figure 37: Full model event probability predictions for Cambodia, Thailand, Laos and Vietnam

The plot value measures the yearly probability that a disaster will happen on each specific point conditional on the occurrence of at least one disaster.



We now zoom in to examine in more detail the specific results for Laos. Figure 38 presents the results for the country. Once again, we can see how the model is able to identify which rivers represent greater risks according to the geospatial patterns of past disasters.

Figure 38: Full model event probability predictions for Lao

The plot value measures the yearly probability that a disaster will happen on each specific point conditional on the occurrence of at least one disaster.

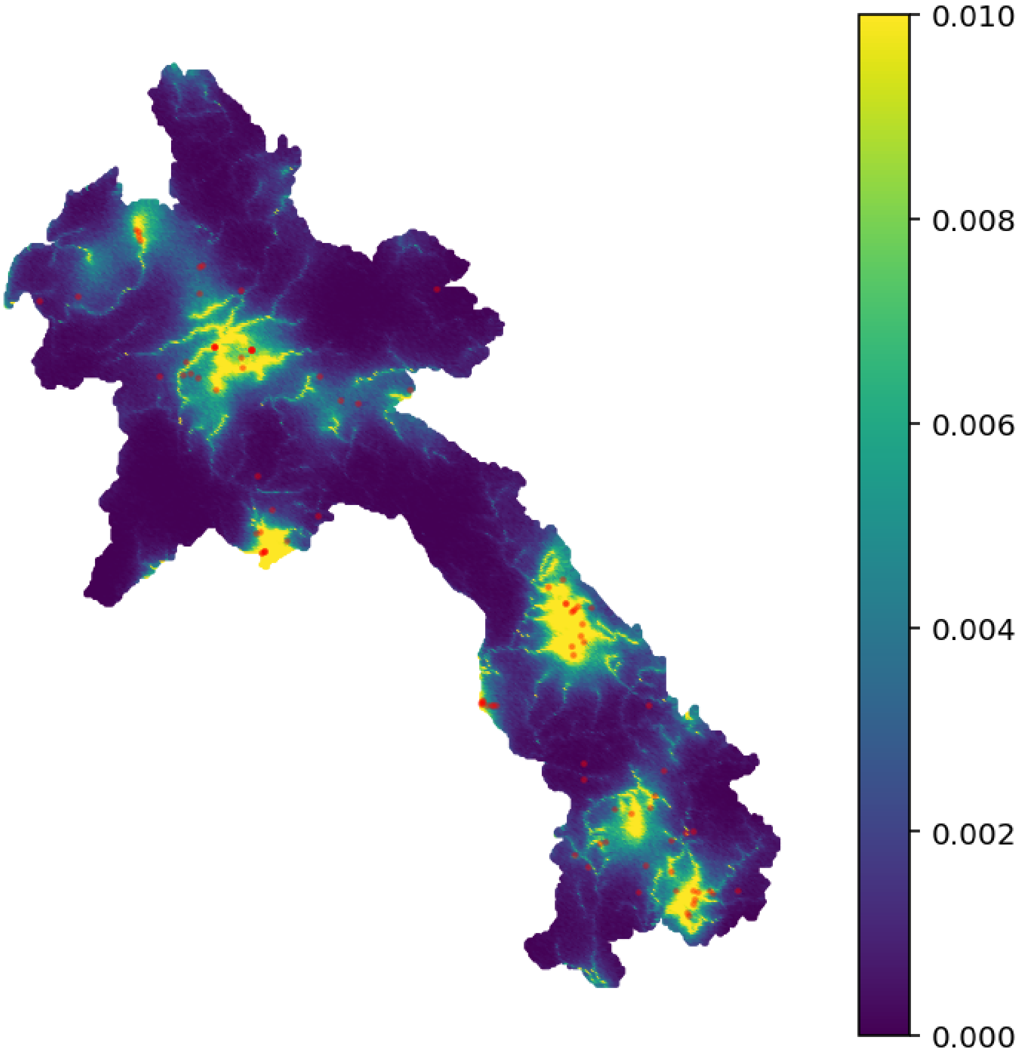
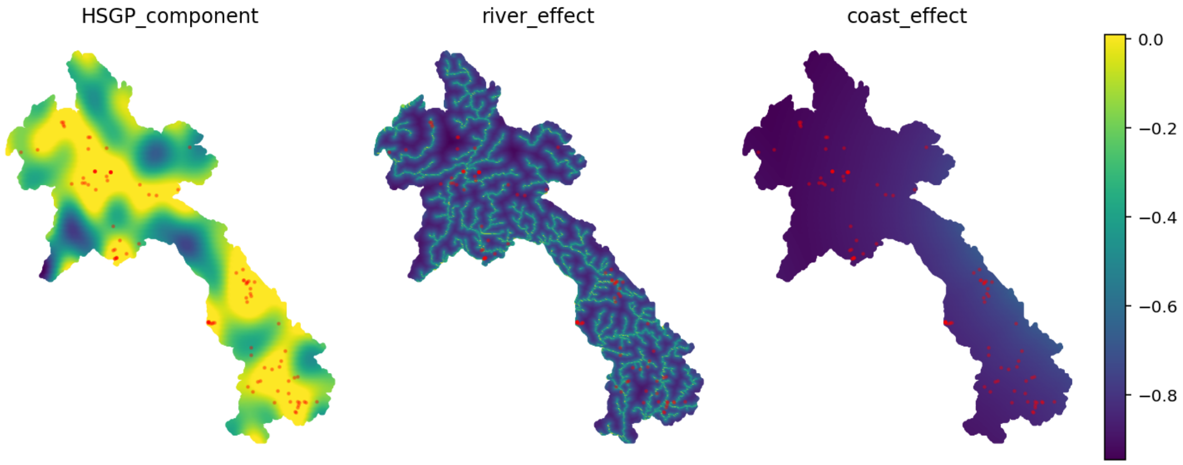


Figure 39 presents the decomposition of the effects for Lao. It is important to consider that besides the effects plotted, the event probability risk is also affected by the country effect, the climate covariates and the country-level covariates, which do not vary at the geospatial level. The decomposition of the effects allows to identify that the HSGP component dominates the projected probability values. It also shows that despite having a lower weight, the distance to coastline has some positive impact, as can be seen in the lower right segment.

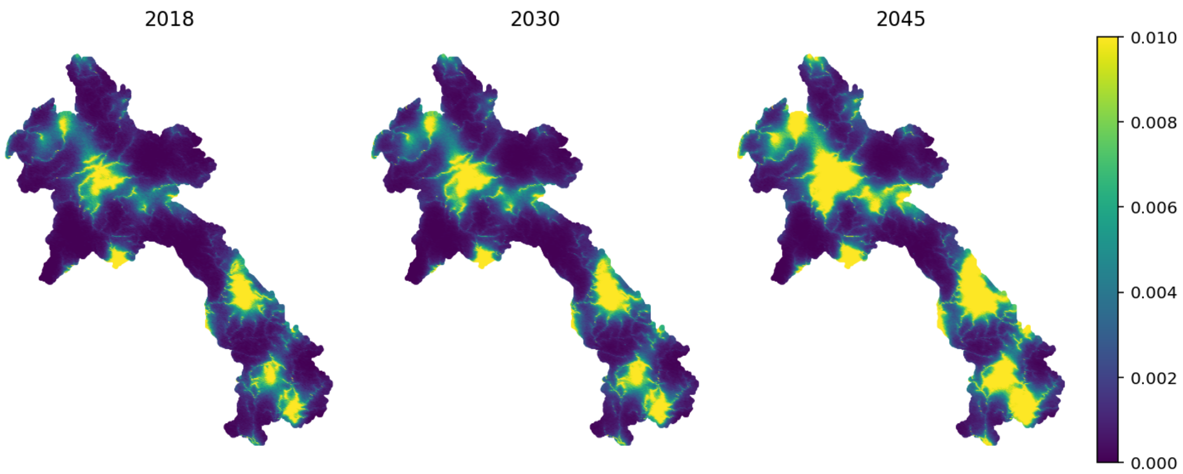
The static geospatial perspective is highly valuable in itself; nevertheless, we are interested in understanding how exposure to climate change risks evolves over time and how it is affected by projected climate trends. To do this, we multiply the geospatial event probability by the mean number of events the country is projected to experience per year. This provides the total event probability per year for each point. We then project these values over time to understand how

Figure 39: Decomposition of the full model event probability predictions for Lao



risk evolves as atmospheric CO_2 increases. Figure 40 shows how local exposure increases over time for Laos.

Figure 40: Full model event probability predictions for Lao through time



We now present the results for Central America and Costa Rica. Figure 41 shows the results for the whole region, while Figure 42 presents the estimations specifically for Costa Rica. In the case of Costa Rica, we observe how the riskiest areas seem to be concentrated in the great metropolitan area, which contains most of the country's population.

Figure 41: Full model event probability predictions for Central America

The plot value measures the yearly probability that a disaster will happen on each specific point conditional on the occurrence of at least one disaster.

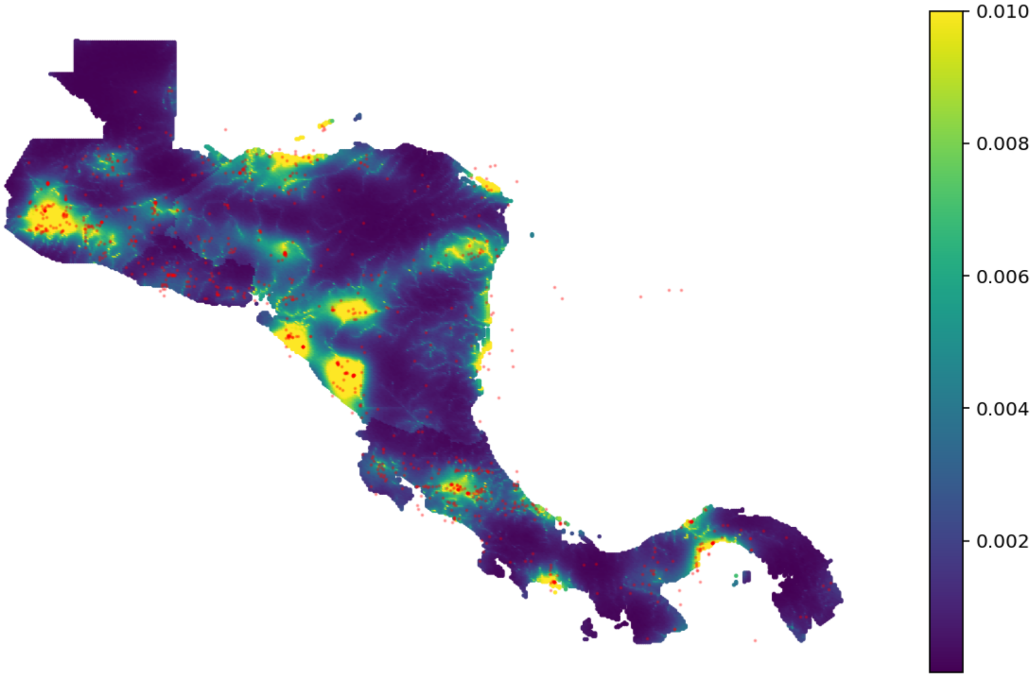


Figure 43 shows the decomposition of the effects. In the case of Costa Rica, we observe the particularity that distance to the coastline has a very limited influence on event probability, with shorter distances to the coastline leading to slight decreases in event probability. One possible explanation for this is the concentration of recorded disasters in more densely populated areas (such as the Greater Metropolitan Area in the center of the country), which results in a lower estimated probability along the coasts, where population density is much lower.

Figure 42: Full model event probability predictions for Costa Rica

The plot value measures the yearly probability that a disaster will happen on each specific point conditional on the occurrence of at least one disaster.

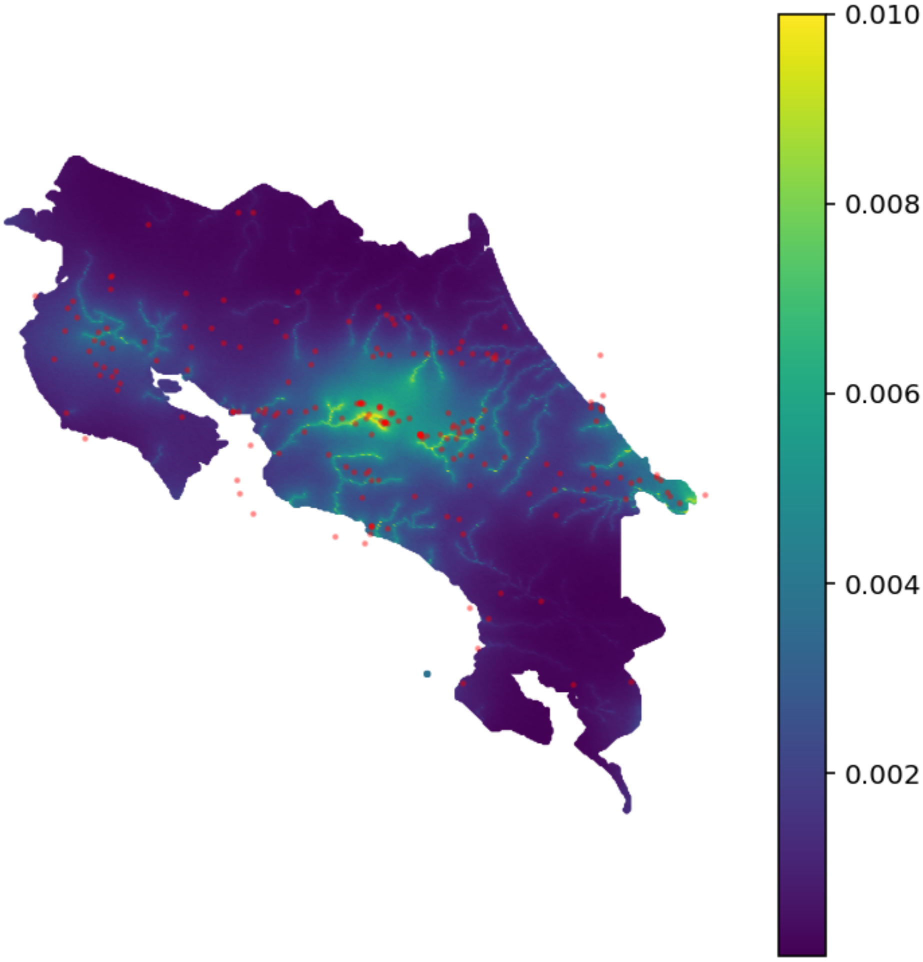
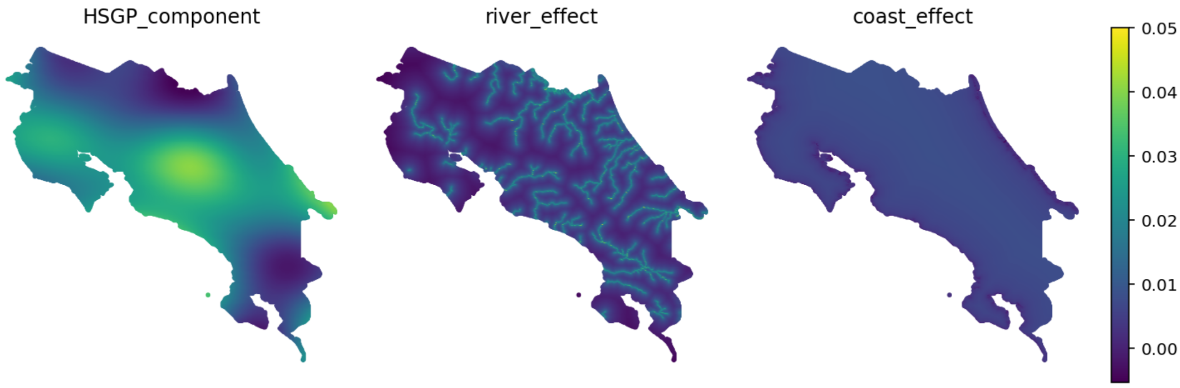
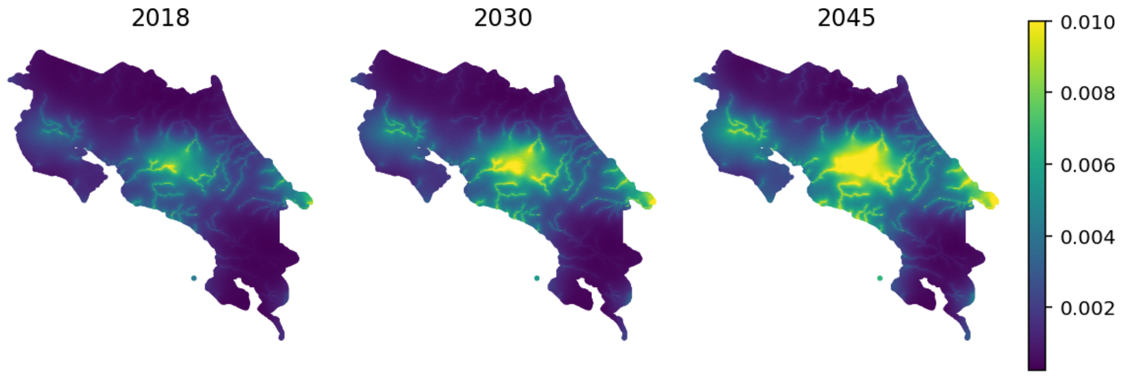


Figure 43: Decomposition of the full model event probability predictions for Costa Rica



To conclude this section, we present on Figure 44 the projected geospatial risk for Costa Rica. Similar to the case of Lao, we can observe how the probability of event increases in the riskier areas as atmospheric CO2 projections grow through time. The projections show a greater exposure for the Greater Metropolitan Area, Guanacaste and the south of the Caribbean.

Figure 44: Full model event probability predictions for Costa Rica through time



6.6 Geolocated return year curves

One of the main contributions of this paper is to provide local-level estimates of event probabilities and the expected damages caused by climate change–related disasters. This approach is not intended to replace on-site hydrological studies, but rather to offer a low-cost and efficient tool that can provide initial approximations of risk.

The model makes it possible to compute these values for any point within the analyzed country. To illustrate this, we focus on three specific locations in Costa Rica:

- Desamparados (San José).
- El Bambú (Filadelfia, Guanacaste).
- Cariari (Pococí, Limón).

These locations are shown in Figure 45.

Figure 45: Selected locations

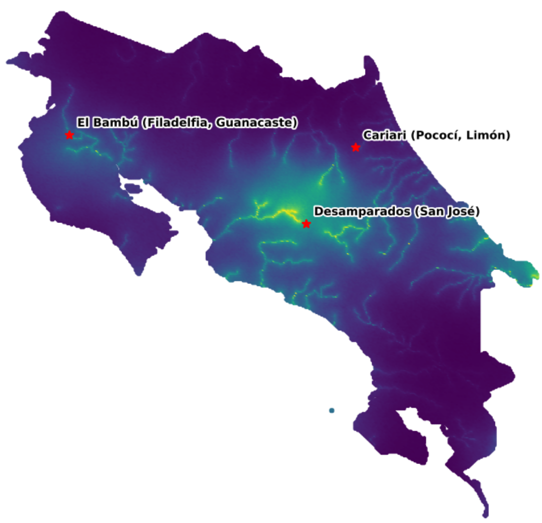


Table 14 shows the yearly probability of an event happening on each point through time. We can observe that Desamparados presents the greater probability and Cariari the lowest one from the three points. Also it is possible to observe how the risk exposure increases for all the locations across time.

Table 14: Projected probability of a disaster on selected locations in Costa Rica through time

The table presents the probability in percentages of one event happening on the selected point at the given year.

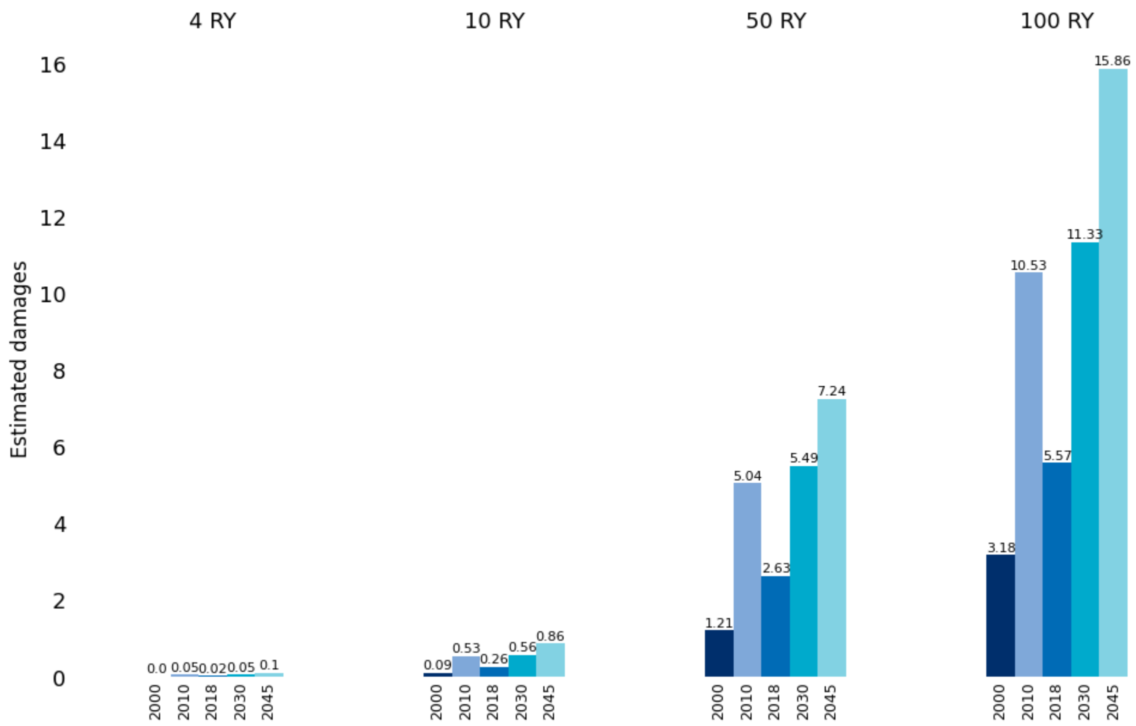
Location	2000	2010	2018	2030	2045
Desamparados (San José)	0.0034	0.0094	0.0065	0.0086	0.0105
El Bambú (Filadelfia, Guanacaste)	0.0019	0.0048	0.0033	0.0050	0.0065
Cariari (Pococí, Limón)	0.0006	0.0017	0.0012	0.0017	0.0027

We use the local probability to scale down the national expected damage, and produce a local level return year curves. These curves do not consider local characteristics of infrastructure, or physical capital, but are local projections of the national damages using the local event probability (which does capture some of this patterns via the HSGP component).

This local projections allows us to quantify the climate change risk exposure at a granular level, and produce approximations of the impacts that particular communities can suffer due to climate change related disasters. To exemplify this, we present on Figure 46 the return year curves for El Bambú (Filadelfia, Guanacaste). Consistent with what has been observed in the previous results, the return year curves reveal a growing pattern in terms of damage exposures as more severe events become more frequent, and more frequent events become more severe.

Figure 46: Geolocated projected return year curves for El Bambú (Filadelfia, Guanacaste)

Estimated damages are measured in millions of 2025 USD.



7 Conclusions

This study introduces the Probabilistic Geospatial Risk (PGR) model, an end-to-end framework designed to quantify and project the frequency, economic damage, and spatial distribution of climate change-related disasters. By integrating Bayesian generative modeling, state-of-the-art spatial analysis, and hierarchical time-series forecasting, the PGR model transcends traditional national-level risk assessments. It provides granular, point-level estimates of disaster risk, enabling policymakers, researchers, and practitioners to identify high-risk areas and prioritize interventions.

The model's ability to capture non-linear relationships and spatial autocorrelation through techniques like the Hilbert Space Gaussian Process (HSGP) and Intrinsic Conditional Autoregressive (ICAR) models provides valuable insights, while its use of generally available satellite information allows predictions even in data-scarce regions.

From Global Trends to Local Action

The PGR model's outputs are highly actionable for climate adaptation and disaster risk reduction. By projecting return-year damage curves and geolocated risk maps, the model bridges the gap between global climate trends and local vulnerability. For example:

Laos and Costa Rica were used as case studies to demonstrate how the model can identify regions with escalating exposure to hydrometeorological disasters. The results reveal that both countries face increasing frequencies of extreme events, with Costa Rica experiencing a sharper rise in projected damages due to its economic and geographic vulnerability.

The geolocated return-year curves for specific locations (e.g., Desamparados, El Bambú, and Cariari in Costa Rica) illustrate how communities can use these tools to assess their unique risks and plan targeted adaptations, such as infrastructure upgrades or early warning systems.

The model's ability to disaggregate national-level risks into local probabilities empowers governments to allocate resources more effectively, focusing on areas where climate impacts are projected to be most severe.

A core innovation of the PGR model is its emphasis on uncertainty quantification. Traditional risk assessments often focus on point estimates (e.g., mean or mode), which are insufficient for policy planning. In contrast, the PGR model leverages posterior predictive distributions to simulate the full range of possible outcomes, including tail risks. This approach is critical for understanding the potential for catastrophic events, which, while rare, can have outsized economic and humanitarian consequences.

The use of Markov Chain Monte Carlo (MCMC) sampling and Bayesian hierarchical modeling ensures that uncertainty is propagated through all stages of the analysis, from frequency and damage modeling to spatial risk mapping. This transparency allows decision-makers to evaluate not just the most likely scenarios but also the worst-case possibilities, fostering more resilient planning.

The study's projections underscore the urgent need for climate action. Under business-as-usual scenarios, both Laos and Costa Rica are expected to experience:

- Higher frequencies of hydrometeorological disasters, with return periods for extreme events shortening dramatically (e.g., 100-year events becoming 50-year events by mid-century).
- Increased severity of damages, as rising ocean temperatures and CO_2 concentrations amplify the intensity of storms, floods, and droughts.
- Spatial shifts in vulnerability, with urban centers (e.g., Costa Rica's Greater Metropolitan Area) and regions close to rivers facing disproportionate risks due to population density and geographic exposure.

These findings align with the broader scientific consensus that climate change will exacerbate disaster risks, but the PGR model provides the granularity needed to translate global trends into local action.

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Annex 1

Figure 47: Kernel density and histogram of the geo-spatial model data set for Costa Rica, El Salvador, Guatemala, Honduras, Nicaragua and Panama

All features, except for “is_disaster” have been standardized.

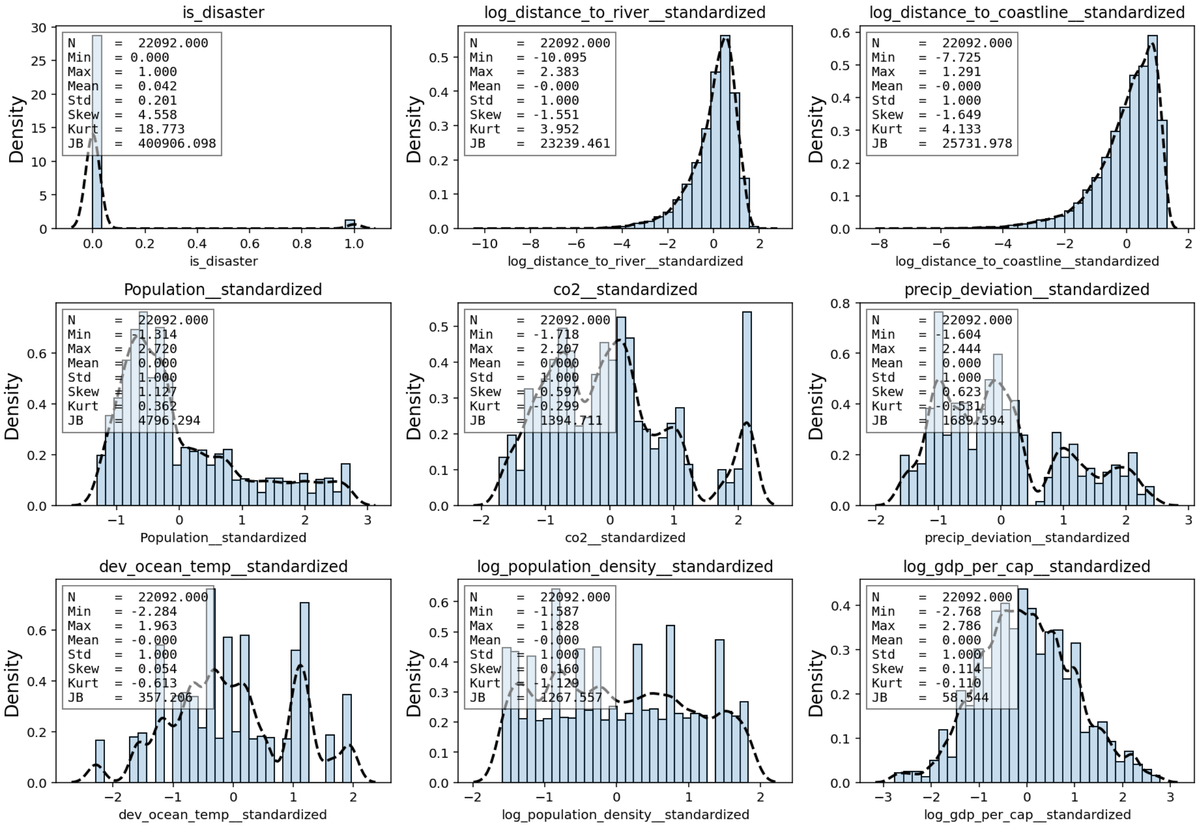


Figure 48: Kernel density and histogram of the geo-spatial model data set for Cambodia, Thailand, Laos and Vietnam

All features, except for “is_disaster” have been standardized.

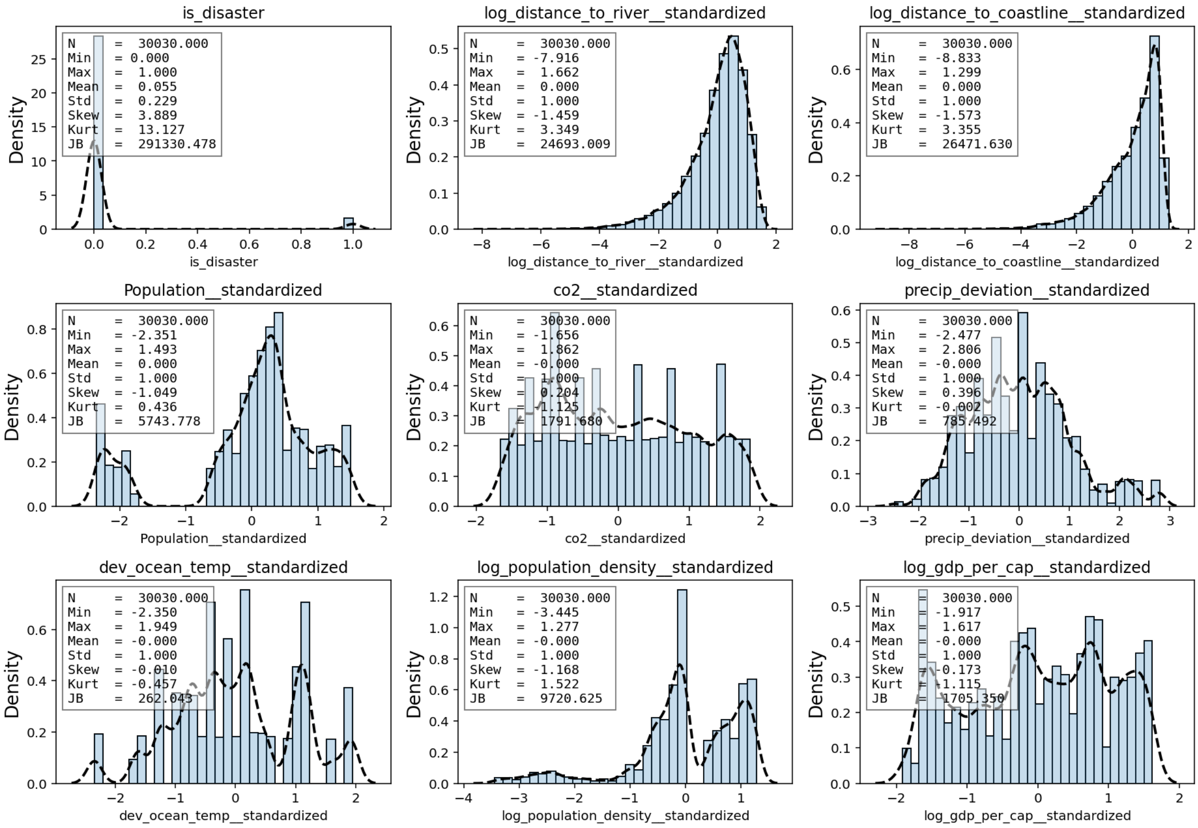


Table 15: Descriptive Statistics of time series

This table presents descriptive statistics for climate time series included in the model. The series have been standardized, and then their first difference was computed.

Statistic	gdp_CRI	Pop_CRI	prec_CRI	gdp_LAO	Pop_LAO	prec_LAO	co2	Temp
Count	44.00	44.00	40.00	40.00	44.00	40.00	45.00	45.00
Mean	-0.00	-0.00	0.00	-0.00	0.00	0.00	-0.00	0.00
Std Dev	1.01	1.01	1.01	1.01	1.01	1.01	1.01	1.01
Min	-1.77	-2.03	-1.50	-1.64	-1.95	-1.92	-1.57	-1.53
25%	-0.85	-0.79	-0.81	-0.70	-0.77	-0.71	-0.83	-0.94
Median	-0.02	0.24	-0.07	-0.39	0.17	0.02	-0.10	-0.10
75%	1.02	0.88	0.72	1.11	0.83	0.93	0.82	0.73
Max	1.58	1.26	1.94	1.50	1.47	1.68	1.85	1.95
Kurtosis	-1.23	-0.99	-0.97	-1.43	-0.99	-0.97	-1.15	-1.11
Skewness	-0.15	-0.52	0.32	0.27	-0.40	-0.21	0.20	0.30

Figure 49: Kernel density and histogram of the time series data

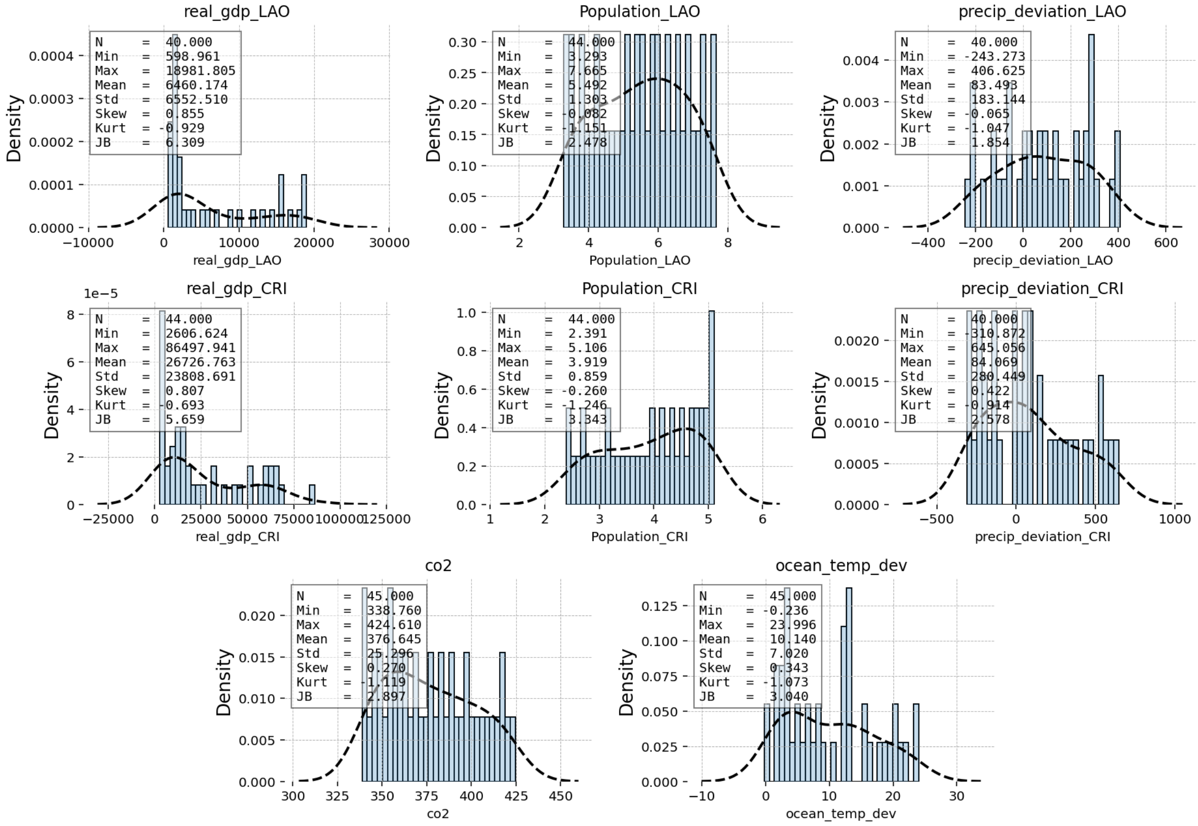
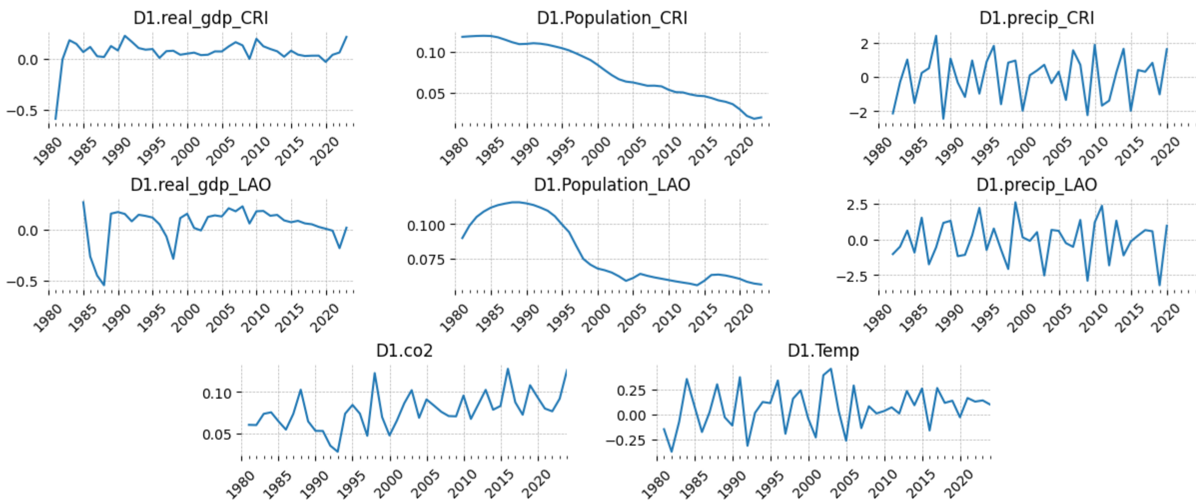


Figure 50: Time series evolution through time

Population is measured in millions, real GDP in millions of USD, CO2 in Gigatons, ocean temperature deviation from its mean in Fahrenheit degrees, and precipitation deviation is the whole year monthly mean in millimeters.



Annex 2

Table 16: Summary Statistics of Model Parameters for HSGP estimation in Cambodia, Thailand, Laos and Vietnam

Parameter	mean	sd	ess_bulk	ess_tail	r_hat
ell_log__[0]	-0.54	0.10	2027.00	2569.00	1.00
ell_log__[1]	-0.62	0.08	2375.00	3180.00	1.00
eta	4.99	0.36	1376.00	2432.00	1.00
eta_log__	1.60	0.07	1376.00	2432.00	1.00

Table 17: Summary Statistics of Model Parameters for HSGP estimation in Central America

Parameter	mean	sd	ess_bulk	ess_tail	r_hat
ell_log__[0]	-0.40	0.11	758.00	1540.00	1.01
ell_log__[1]	-0.09	0.11	673.00	1566.00	1.01
eta	3.81	0.40	540.00	1223.00	1.01
eta_log__	1.33	0.10	540.00	1223.00	1.01

Table 18: Summary Statistics of Full Model Parameters for Cambodia, Thailand, Laos and Vietnam

Parameter	mean	sd	ess_bulk	ess_tail	r_hat
ell_log__[0]	-1.37	0.25	3987.00	4003.00	1.00
ell_log__[1]	-1.35	0.18	4197.00	4835.00	1.00
eta	3.12	0.36	2088.00	3751.00	1.00
eta_log__	1.13	0.11	2088.00	3751.00	1.00
beta[log_distance_to_river__standardized]	-0.44	0.03	11561.00	5363.00	1.00
beta[log_distance_to_coastline__standardized]	-0.17	0.06	7776.00	5073.00	1.00
beta[Population__standardized]	0.10	0.09	14518.00	6600.00	1.00
beta[co2__standardized]	0.25	0.06	13074.00	6135.00	1.00
beta[precip_deviation__standardized]	0.33	0.03	8833.00	5334.00	1.00
beta[dev_ocean_temp__standardized]	0.02	0.03	8344.00	4750.00	1.00
beta[log_population_density__standardized]	0.32	0.09	14313.00	6382.00	1.00
beta[log_gdp_per_cap__standardized]	0.12	0.08	15245.00	6216.00	1.00
country_effect[KHM]	-1.75	0.41	5900.00	5188.00	1.00
country_effect[LAO]	-1.73	0.43	7930.00	6514.00	1.00
country_effect[THA]	-4.52	0.26	6296.00	5787.00	1.00
country_effect[VNM]	-3.87	0.30	5890.00	5449.00	1.00

Table 19: **Summary Statistics of Full Model Parameters for Central America**

Parameter	mean	sd	ess_bulk	ess_tail	r_hat
ell_log__[0]	-0.48	0.12	4541.00	5254.00	1.00
ell_log__[1]	-0.22	0.14	3912.00	5530.00	1.00
eta	3.34	0.43	3177.00	4311.00	1.00
eta_log__	1.20	0.13	3177.00	4311.00	1.00
beta[log_distance_to_river__standardized]	-0.25	0.05	10393.00	6385.00	1.00
beta[log_distance_to_coastline__standardized]	0.02	0.06	10822.00	6151.00	1.00
beta[Population__standardized]	0.25	0.08	10462.00	6294.00	1.00
beta[co2__standardized]	0.44	0.05	10143.00	6889.00	1.00
beta[precip_deviation__standardized]	0.45	0.03	9750.00	5208.00	1.00
beta[dev_ocean_temp__standardized]	-0.19	0.04	9240.00	5748.00	1.00
beta[log_population_density__standardized]	0.27	0.09	11022.00	6220.00	1.00
beta[log_gdp_per_cap__standardized]	0.08	0.09	11172.00	6233.00	1.00
country_effect[CRI]	-1.30	0.64	5644.00	5761.00	1.00
country_effect[GTM]	-1.80	0.68	5179.00	5044.00	1.00
country_effect[HND]	-0.05	0.59	5167.00	5114.00	1.00
country_effect[NIC]	-0.53	0.62	5291.00	5985.00	1.00
country_effect[PAN]	-0.91	0.69	5392.00	5844.00	1.00
country_effect[SLV]	-1.68	0.67	6221.00	6578.00	1.00